

Gene Set Analysis

HUGEN 2073

Jon Chernus

What can you do with a list of genes?

- We can do “variant → gene” inference at each locus, using
 - LD/position
 - Known trait associations
 - eQTLS (in a later lecture)
 - Etc.
- Put all the loci together → a list of genes

How can you interpret a list of genes?

Suppose your study (e.g., GWAS) yielded a list of dozens (or more) genes

- Why not just publish a list of genes and stop there?
 - Are all those genes necessarily relevant?
 - What scientific/statistical question(s) can you ask about the list?

Gene set analysis

- “Are certain categories of genes “enriched” in my results, compared to what would be expected by chance?”
- Originally used for differential expression studies

Includes or can be sometimes synonymous with

- “enrichment analysis”
- “over-representation analysis”
- Etc.

Some considerations

- What is the null hypothesis of a given method?
- Some methods can include effect sizes and directions (instead of simply using gene names)

Over-representation/enrichment

When asking what categories of genes are over-represented/enriched among a prioritized list of genes, what kinds of categories exist?

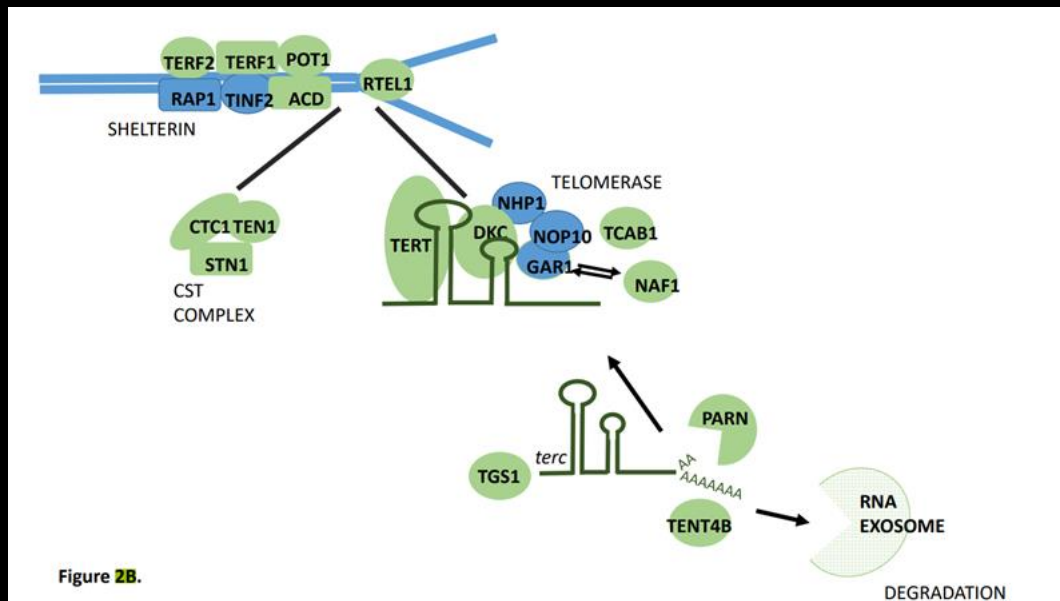
- What kinds of categories might you be interested in?

Over-representation

Example: a GWAS of telomere length

<https://www.medrxiv.org/content/10.1101/2021.03.23.21253516v1>

Key components of telomere regulatory complexes found within genome-wide significant loci.



Proteins depicted in green are found within GWAS loci, those not found within GWAS loci are depicted in blue.

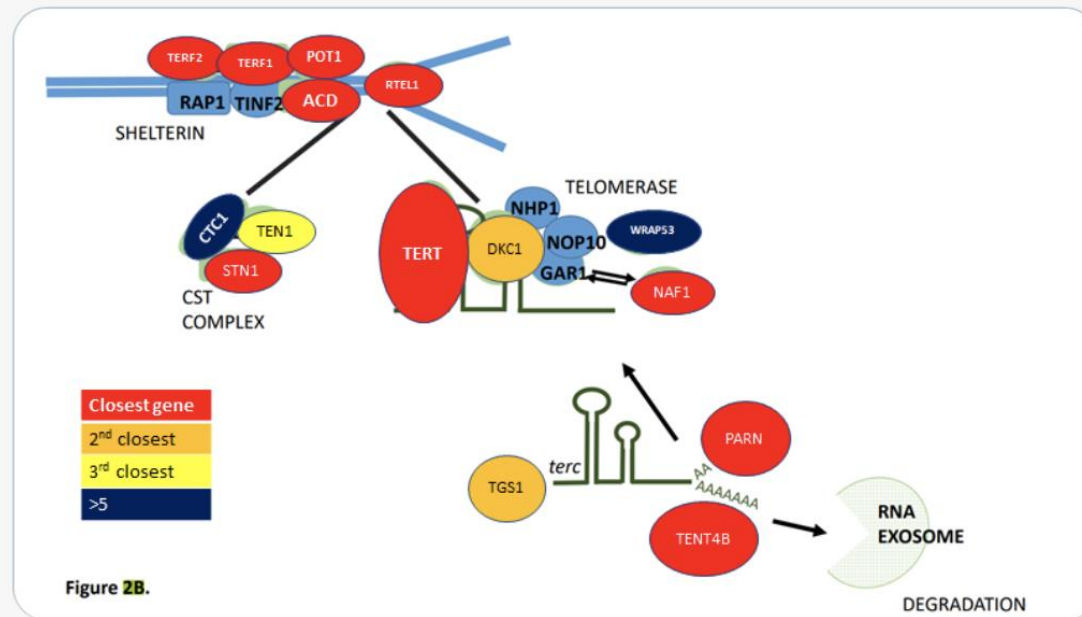
We find the majority of components of core telomere binding complexes alongside many proteins involved in the formation and activity of telomerase. Note that not all components of the RNA exosome are shown.

Over-representation



Eric Fauman @Eric_Fauman · Mar 26

Here I colored each gene by its closest proximity to a GWAS SNP. Genes in red represent a closest gene to a GWAS SNP. Orange means 1 other gene is closer. TEN1 (yellow) is the 3rd closest gene to its SNP. And 2 genes are further than 5th closest gene. Pretty amazing I think!



It looks like “telomerase-maintenance” genes are probably over-represented in these GWAS results.

(In a GWAS of telomere length, this is not surprising.)

But how to quantify the over-representation?

Some examples of gene sets/databases/tools

Some databases

- Gene Ontologies (GO) (<http://geneontology.org>)
- KEGG (<https://www.genome.jp/kegg/>)
- Reactome (<https://reactome.org>)
- MSigDB
- GTEx (tissue-specific gene sets)
- Pathway Commons (<https://www.pathwaycommons.org>)
- Many others

Some methods and implementations

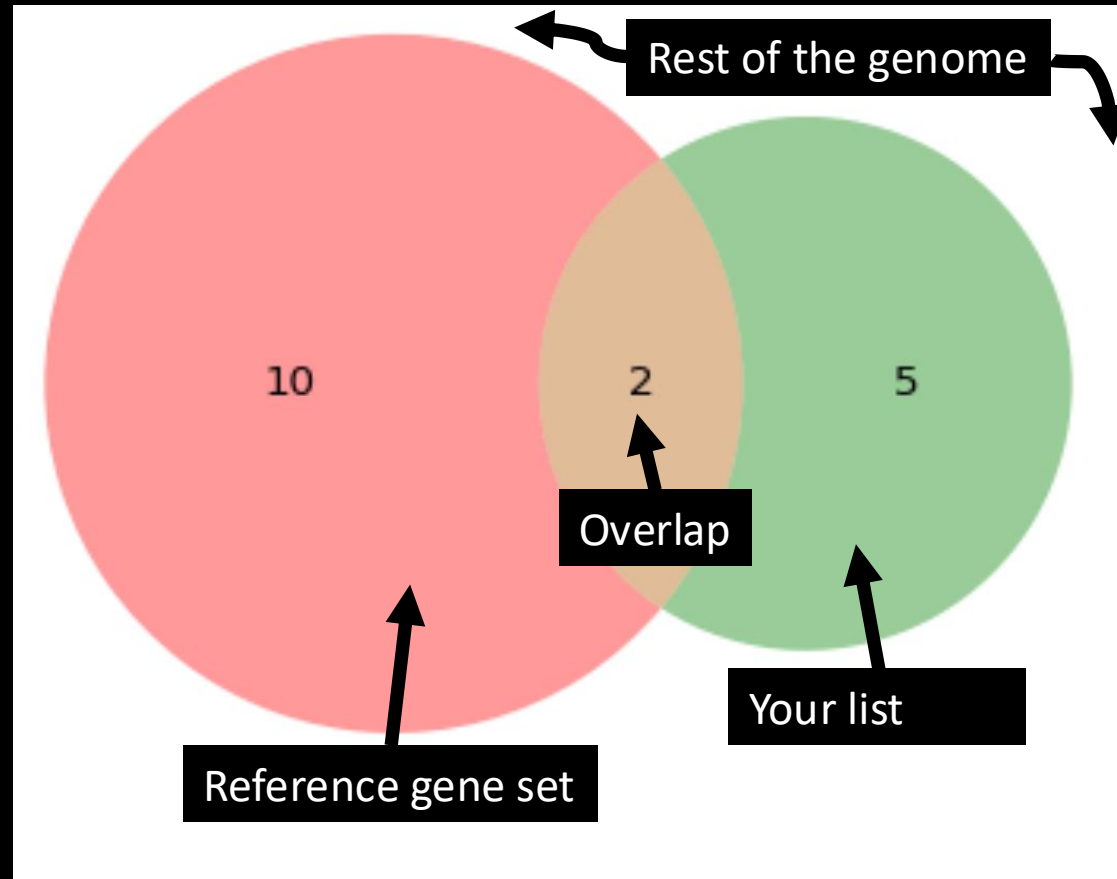
- PANTHER (<http://www.pantherdb.org>)
- IPA (<https://digitalinsights.qiagen.com/products-overview/discovery-insights-portfolio/analysis-and-visualization/qiagen-ipa/>; not free)
- GREAT (<http://great.stanford.edu/public/html/>)
- GSEA (<https://www.gsea-msigdb.org/gsea/index.jsp>)
- DAVID (<https://david.ncifcrf.gov>)
- MAGMA (<https://ctg.cncr.nl/software/magma>)
- FUMA (GENE2FUNC function; MAGMA) (<https://fuma.ctglab.nl>)
- Enrichr (<https://maayanlab.cloud/Enrichr/>)
- STRING
- R packages: fgsea, clusterProfiler

Types of gene set analysis

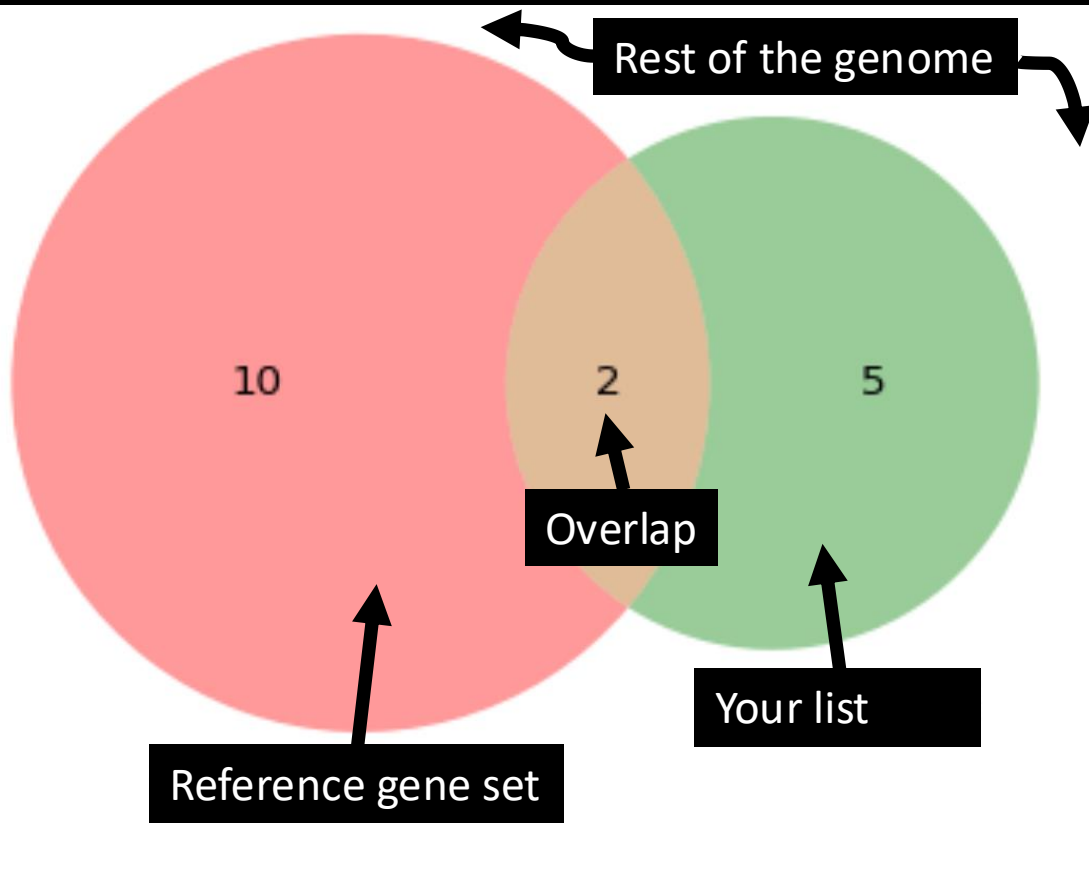
- Over-representation analysis (think Fisher's exact test)
 - Simplest
 - You have a thresholded gene list
 - Compare overlap with a gene set
 - Interpretation: "More hits in this pathway than expected by chance"
- Functional class scoring (think GSEA)
 - Uses all genes you have results for, ranked (not thresholded)
 - Interpretation: "This pathway has *stronger* hits than expected by chance"
- Pathway topology-based (e.g., SPIA)
 - Use pathway structure
 - Weigh genes by position/connectivity
 - Interpretation: "Important parts of this pathway show signal"

The simplest over-representation test

Venn diagram to show enrichment



Hypergeometric test (Fisher's exact test)



What's the probability of an overlap at least this big?

- You picked $n=7$ GWAS genes out of the genome ($N \approx 20,000$)
- The gene set had $K=10$ genes, overlapping $k=2$ of your genes

Number of ways to pick b balls from an urn with a total balls:

$${}_a C_b = \binom{a}{b} = \frac{a!}{(a-b)!(b!)}$$

How many ways could you pick 7 genes out of the genome?

$${}_N C_n = \binom{20000}{7} = \frac{20000!}{(20000-7)!(7!)} = ?$$

How many ways could you get an overlap of exactly 2?

$${}_K C_k * {}_{N-K} C_{n-k} = \binom{10}{2} * \binom{20,000-10}{5} = \frac{10!}{(10-2)!(2!)} * \frac{(20,000-10)!}{((20,000-10)-5)!(5!)} = ?$$

So $P[\text{overlap exactly } k=2] = ? / ?$

$$\text{Finally, } P_{\text{hypergeometric}} = P[\text{see overlap at least this big}] = \sum_{i=k}^{\min(K,n)} P(\text{overlap exactly } i)$$

Gene Ontologies

<http://geneontology.org>

Gene ontology (GO) is a formalized system for categorizing/annotating genes according to their biological functions.

This makes it easier to systematically compare gene sets and also can help keep track of similar genes across different organisms.

Every gene has assigned three “aspects”

- Molecular function
- Cellular component
- Biological process

Gene Ontologies - aspects

Molecular Function	Molecular-level activities performed by gene products. Molecular function terms describe activities that occur at the molecular level, such as “catalysis” or “transport”. GO molecular function terms represent activities rather than the entities (molecules or complexes) that perform the actions, and do not specify where, when, or in what context the action takes place. Molecular functions generally correspond to activities that can be performed by individual gene products (<i>i.e.</i> a protein or RNA), but some activities are performed by molecular complexes composed of multiple gene products. Examples of broad functional terms are <i>catalytic activity</i> and <i>transporter activity</i> ; examples of narrower functional terms are <i>adenylate cyclase activity</i> or <i>Toll-like receptor binding</i> . To avoid confusion between gene product names and their molecular functions, GO molecular functions are often appended with the word “activity” (a <i>protein kinase</i> would have the GO molecular function <i>protein kinase activity</i>).
Cellular Component	The locations relative to cellular structures in which a gene product performs a function, either cellular compartments (<i>e.g.</i> , <i>mitochondrion</i>), or stable macromolecular complexes of which they are parts (<i>e.g.</i> , the <i>ribosome</i>). Unlike the other aspects of GO, cellular component classes refer not to processes but rather a cellular anatomy.
Biological Process	The larger processes, or ‘biological programs’ accomplished by multiple molecular activities. Examples of broad biological process terms are <i>DNA repair</i> or <i>signal transduction</i> . Examples of more specific terms are <i>pyrimidine nucleobase biosynthetic process</i> or <i>glucose transmembrane transport</i> . Note that a biological process is not equivalent to a pathway. At present, the GO does not try to represent the dynamics or dependencies that would be required to fully describe a pathway.

In an example of GO annotation, the gene product “cytochrome c” can be described by the **molecular function** *oxidoreductase activity*, the **biological process** *oxidative phosphorylation*, and the **cellular component** *mitochondrial matrix*.

The GO vocabulary is designed to be species-agnostic, and includes terms applicable to prokaryotes and eukaryotes, as well as single and multicellular organisms.

Gene Ontologies – GO terms

Each possible aspect is arranged in a hierarchy or “tree” ranging from the most general to the most specific.

If a gene belongs to a specific class, it also belongs to all of the “parent” classes in the hierarchy containing that class.

The entire system of classes can be described with a formalized logical vocabulary.

Gene Ontologies – GO terms

General principles of GO annotations

- Annotations represent the normal functions of gene products.
- A gene product can be annotated to zero or more terms from each ontology.
- Each annotation is supported by an [GO Evidence Codes](#) from the [Evidence and Conclusions Ontology](#) and a reference.
- Gene products are annotated to the most granular term in the ontology that is supported by the available evidence.
- By the transitivity principle, an annotation to a GO term implies annotation to all its parents.
- GO annotations are meant to reflect the most up-to-date view of a gene product's role in biology.
- Because biological knowledge changes, annotations for a given gene product may change to reflect changes in knowledge and/or changes in the ontology.
- There is an open-world assumption, that is, if a gene product is unannotated then its role is still unknown.

Gene Ontologies – GO terms

- The possible values for a given aspect are called GO terms or classes and have unique identifiers and descriptions:

Unique identifier and term name

Every term has a human-readable term name — e.g. [mitochondrion](#), [glucose transmembrane transport](#), or [amino acid binding](#) — and a GO ID, a unique seven digit identifier prefixed by GO:, e.g. [GO:0005739](#), [GO:1904659](#), or [GO:0016597](#).

Gene Ontologies

Searching a gene

**GENE ONTOLOGY**
Unifying Biology

AboutOntologyAnnotationsDownloadsHelp



 COVID-19 pandemic: click here to get GO data on SARS-CoV-2

Current release 2021-02-01: 44,085 GO terms | 7,931,218 annotations
1,564,454 gene products | 4,743 species ([see statistics](#))

THE GENE ONTOLOGY RESOURCE

The mission of the GO Consortium is to develop a comprehensive, **computational model of biological systems**, ranging from the molecular to the organism level, across the multiplicity of species in the tree of life.

The Gene Ontology (GO) knowledgebase is the world's largest source of information on the functions of genes. This knowledge is both human-readable and machine-readable, and is a foundation for computational analysis of large-scale molecular biology and genetics experiments in biomedical research.

SLC39A10

Any ● Ontology ● Gene Product

GO Enrichment Analysis ?

Powered by PANTHER

Your gene IDs here...

biological process

Homo sapiens

ExamplesLaunch >

Hint: can use UniProt ID/AC, Gene Name, Gene Symbols, MOD IDs

Gene Ontologies

Search results. Let's click on "genes and gene products."

 AmiGO 2 Home Search ▾ Browse Tools & Resources Help Feedback About



Text search document selection 

The following results were found for **SLC39A10** using a general search over all text fields.

To narrow your search, select the type of document that you would like to search for and continue narrowing your search from the linked search page.

Ontology	Gene Ontology Term, Synonym, or Definition.	0
<input checked="" type="button" value="Genes and gene products"/>	Genes and gene products associated with GO terms.	18
Annotations	Associations between GO terms and genes or gene products.	133

Gene Ontologies

Let's select Homo sapiens.

AmiGO 2

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Search

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Tools & Resources

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Feedback

About

Quick search

Search

Information about Genes and gene products search

Filter results

Total gene product(s): 18

SLC39A10

No current user filters.

Your search is pinned to these filters

- document_category: bioentity

Source

Organism

Type

PANTHER family

Direct annotation

Inferred annotation

Total gene product(s): 18; showing: 1-10

Results count 10

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Bookmark

Gene/product	Gene/product name	Organism	PANTHER family	Type	Source	Synonyms
☐ SLC39A10	Uncharacterized protein	Felis catus	solute carrier family 39 pthr12191	protein	UniProtKB	UniProtKB:A0A2I2UF00 PTN001742834
☐ SLC39A10	SLC39A10 isoform 2	Pan troglodytes	solute carrier family 39 pthr12191	protein	UniProtKB	UniProtKB:A0A2I3TSU1 PTN000257395
☐ slc39a10	Solute carrier family 39 member 10	Xenopus tropicalis	solute carrier family 39 pthr12191	protein	Xenbase	UniProtKB:F6T9T9 PTN000257409
☐ SLC39A10	Uncharacterized protein	Monodelphis domestica	solute carrier family 39 pthr12191	protein	UniProtKB	UniProtKB:F7ALF5 PTN000257407
☐ SLC39A10	Uncharacterized protein	Canis lupus familiaris	solute carrier family 39 pthr12191	protein	UniProtKB	UniProtKB:F1PA89 PTN000257406
☐ SLC39A10	Solute carrier family 39 member 10	Canis lupus familiaris		protein	UniProtKB	
☐ SLC39A10	Zinc transporter ZIP10	Homo sapiens	solute carrier family 39 pthr12191	protein	UniProtKB	KIAA1265 ZIP10
☐ SLC39A10	Solute carrier family 39 member 10	Sus scrofa		protein	UniProtKB	
☐ Slc39a10	solute carrier family 39 member 10	Rattus norvegicus	solute carrier family 39 pthr12191	gene	RGD	
☐ slc39a10	solute carrier family 39 member 10	Danio rerio	solute carrier family 39 pthr12191	protein_coding_gene	ZFIN	zgc:63552 ZIP10



SLC39A10

Zinc transporter ZIP10

Homo sapiens

solute carrier family
39 pthr12191

protein

UniProtKB KIAA1265
ZIP10

Gene Ontologies

Results for *SLC39A10*

Zinc transporter ZIP10

Gene Product Information

Symbol SLC39A10

Name(s) Zinc transporter ZIP10

Type protein

Taxon Homo sapiens

Synonyms KIAA1265
ZIP10

Database UniProtKB, Q9ULF5

Related [Link](#) to annotations page to SLC39A10.
[Link](#) to annotations download for SLC39A10.

Data health

Gene Product Associations

Filter results

Total annotations: 8

No current user filters.

Your search is pinned to these filters

- document_category: annotation
- bioentity: UniProtKB:Q9ULF5

Ontology (aspect)

+	-	(6)	P
+	-	(1)	C
+	-	(1)	F

Organism

Type

Evidence

GO class

GO class (direct)

Annotation qualifier

Annotation extension

Contributor

PANTHER family

Total annotations: 8; showing: 1-8

Results count 10

[~First](#) [<Prev](#) [Next>](#) [Last>](#) [Download \(up to 100000\)](#)

Gene/product	Gene/product name	Annotation qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence with	PANTHER family	Type	Isoform	Reference	Date
SLC39A10	Zinc transporter ZIP10		negative regulation of B cell apoptotic process		Ensembl	Homo sapiens	IEA	UniProtKB:Q6P5F6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		positive regulation of B cell proliferation		Ensembl	Homo sapiens	IEA	UniProtKB:Q6P5F6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		positive regulation of protein tyrosine phosphatase activity		Ensembl	Homo sapiens	IEA	UniProtKB:Q6P5F6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		zinc ion import across plasma membrane	GO_Central		Homo sapiens	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A10	Zinc transporter ZIP10		cellular zinc ion homeostasis	GO_Central		Homo sapiens	IBA	PANTHER:PTN000953562 UniProtKB:Q13433 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20170228
SLC39A10	Zinc transporter ZIP10		zinc ion transmembrane transporter activity	GO_Central		Homo sapiens	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20170228
SLC39A10	Zinc transporter ZIP10		positive regulation of B cell receptor signaling pathway	GO_Central		Homo sapiens	IBA	PANTHER:PTN000953562 MGI:MGI:1914515	solute carrier family 39 pthr12191	protein		PMID:21873635	20200317
SLC39A10	Zinc transporter ZIP10		integral component of plasma membrane	GO_Central		Homo sapiens	IBA	PANTHER:PTN000953562 FB:FBgn0024236 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20170228

Gene Ontologies

Results for *SLC39A10*. Filter to show only aspect F (function)

Zinc transporter ZIP10

Gene Product Information

Symbol SLC39A10

Name(s) Zinc transporter ZIP10

Type protein

Taxon Homo sapiens

Synonyms KIAA1265
ZIP10

Database UniProtKB, Q9ULF5

Related [Link](#) to annotations page to SLC39A10.
[Link](#) to annotations download for SLC39A10.

Data health

Gene Product Associations

Filter results

Total annotations: 1

User filters

+ aspect: F

Your search is pinned to these filters

- document_category: annotation

- bioentity: UniProtKB:Q9ULF5

Ontology (aspect)

Nothing to filter.

Organism

Type

Evidence

GO class

GO class (direct)

Annotation qualifier

Annotation extension

Contributor

PANTHER family

Total annotations: 1; showing: 1-1

Results count 10

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Gene/product	Gene/product name	Annotation qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence with	PANTHER family	Type	Isoform	Reference	Date
SLC39A10	Zinc transporter ZIP10		zinc ion transmembrane transporter activity		GO_Central	Homo sapiens	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20170228

Gene Ontologies

Results for *SLC39A10*. Filter to show only aspect P (process)

Zinc transporter ZIP10

Gene Product Information

Symbol SLC39A10

Name(s) Zinc transporter ZIP10

Type protein

Taxon Homo sapiens

Synonyms KIAA1265
ZIP10

Database UniProtKB, [Q9ULF5](#)

Related [Link](#) to annotations page to SLC39A10.
[Link](#) to annotations download for SLC39A10.

Data health

Gene Product Associations

Filter results

Total annotations: 6

User filters

+ aspect: P

Your search is pinned to these filters

- document_category: annotation

- biocounty: UniProtKB:Q9ULF5

Ontology (aspect)

Nothing to filter.

Organism

Type

Evidence

GO class

GO class (direct)

Annotation qualifier

Annotation extension

Contributor

PANTHER family

Total annotations: 6; showing: 1-6

Results count 10

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Gene/product	Gene/product name	Annotation qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence with	PANTHER family	Type	Isoform	Reference	Date
SLC39A10	Zinc transporter ZIP10		negative regulation of B cell apoptotic process		Ensembl	Homo sapiens	IEA	UniProtKB:Q6PSF6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		positive regulation of B cell proliferation		Ensembl	Homo sapiens	IEA	UniProtKB:Q6PSF6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		positive regulation of protein tyrosine phosphatase activity		Ensembl	Homo sapiens	IEA	UniProtKB:Q6PSF6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		zinc ion import across plasma membrane		GO_Central	Homo sapiens	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A10	Zinc transporter ZIP10		cellular zinc ion homeostasis		GO_Central	Homo sapiens	IBA	PANTHER:PTN000953562 UniProtKB:Q13433 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20170228
SLC39A10	Zinc transporter ZIP10		positive regulation of B cell receptor signaling pathway		GO_Central	Homo sapiens	IBA	PANTHER:PTN002638848 MGI:MGI:1914515	solute carrier family 39 pthr12191	protein		PMID:21873635	20200317

Gene Ontologies

Result of clicking on one of the GO terms: “zinc ion import across plasma membrane.”

zinc ion import across plasma membrane

Term Information

Accession

GO:0071578

Name

zinc ion import across plasma membrane

Ontology

biological_process

Synonyms

zinc II ion plasma membrane import, zinc II ion transmembrane import, zinc import, zinc uptake, z...

Alternative IDs

GO:0006631, GO:0140160, GO:0006630, GO:0044749

Definition

The directed movement of zinc(2+) ions from outside of a cell, across the plasma membrane and into the cytosol. Source: GOC:vw, PMID:16637840

Comment

None

History

See term history for GO:0071578 at QuickGO

Subset

None

Related

Link

to all genes and gene products annotated to zinc ion import across plasma membrane.

Link

to all direct and indirect annotations to zinc ion import across plasma membrane.

Link

to all direct and indirect annotations download (limited to first 10,000) for zinc ion import across plasma membrane.

Data health

Annotations

Graph Views

Inferred Tree View

Neighborhood

Mappings

Filter results

Total annotations: 272

No current user filters.

Your search is pinned to these filters

-- document_category: annotation

-- regulates_closure: GO:0071578

Ontology (aspect)

Organism

Type

Evidence

GO class

GO class (direct)

Annotation qualifier

Annotation extension

Contributor

PANTHER family

Total annotations: 272; showing: 1-10

Results count 10 3

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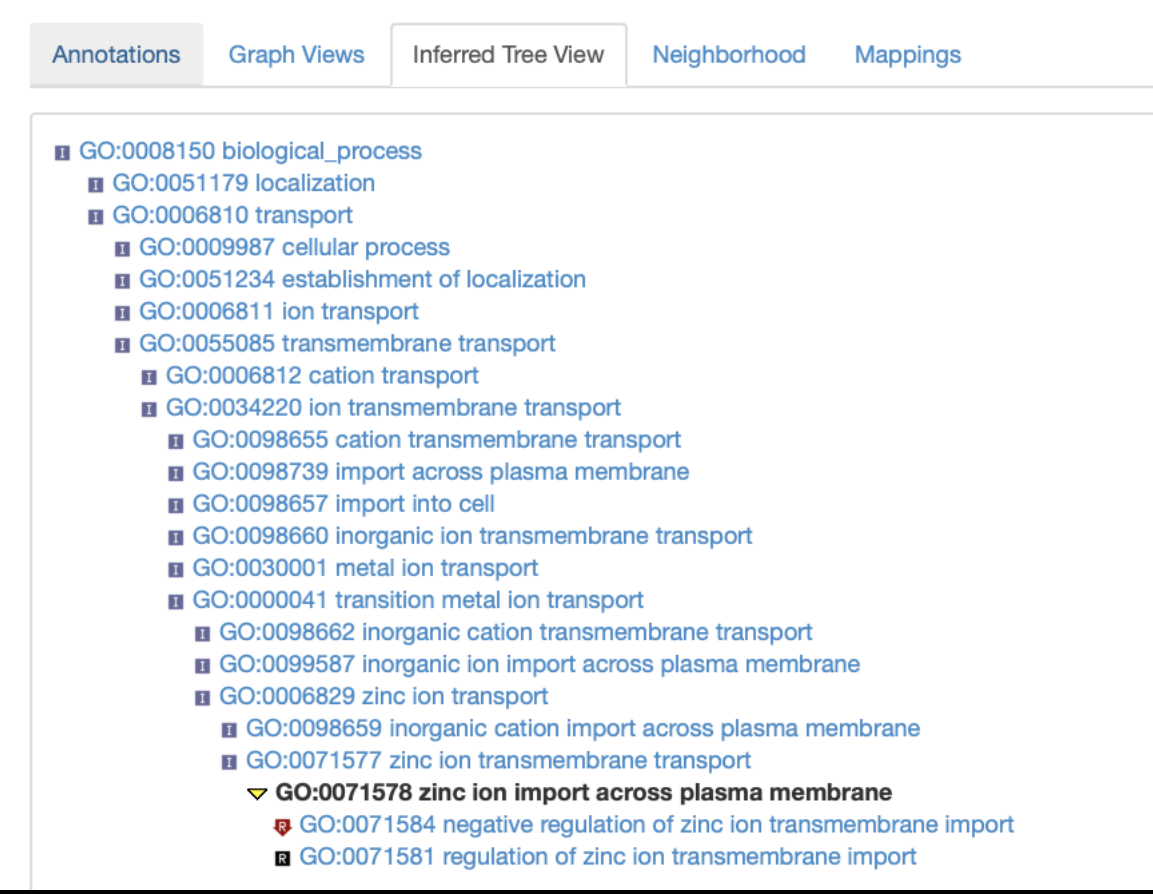
Last>

Download

Gene/product	Gene/product name	Annotation qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence with	PANTHER family	Type	Isoform	Reference	Date
AD212YLD7	Solute carrier family 39 member 14		zinc ion import across plasma membrane		GO_Central	Gorilla gorilla gorilla	IBA	PANTHER:PTN000953562 MG:MG:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
AD212YX43	Solute carrier family 39 member 10		zinc ion import across plasma membrane		GO_Central	Gorilla gorilla gorilla	IBA	PANTHER:PTN000953562 MG:MG:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A14	Uncharacterized protein		zinc ion import across plasma membrane		GO_Central	Felis catus	IBA	PANTHER:PTN000953562 MG:MG:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A6	Solute carrier family 39 member 6		zinc ion import across plasma membrane		GO_Central	Equus caballus	IBA	PANTHER:PTN000953562 MG:MG:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A10	Uncharacterized protein		zinc ion import across plasma membrane		GO_Central	Felis catus	IBA	PANTHER:PTN000953562 MG:MG:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A12	Zinc transporter ZIP12		zinc ion import across plasma membrane		GO_Central	Bos taurus	IBA	PANTHER:PTN000953562 MG:MG:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
XENTR_v90030554mg	Uncharacterized protein		zinc ion import across plasma membrane		GO_Central	Xenopus tropicalis	IBA	PANTHER:PTN000953562 MG:MG:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
BRAF1DRAFT_105599	Uncharacterized protein		zinc ion import across plasma membrane		GO_Central	Branchiostoma floridae	IBA	PANTHER:PTN000953562 MG:MG:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A6	Uncharacterized protein		zinc ion import across plasma membrane		GO_Central	Bos taurus	IBA	PANTHER:PTN000953562 MG:MG:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
v1g85598	Predicted protein (Fragment)		zinc ion import across plasma		GO_Central	Nematostella vectensis	IBA	PANTHER:PTN000953562 MG:MG:1914797 more...	solute carrier family 39	protein		PMID:21873635	20200808

Gene Ontologies

Result of clicking on “Inferred Tree View”



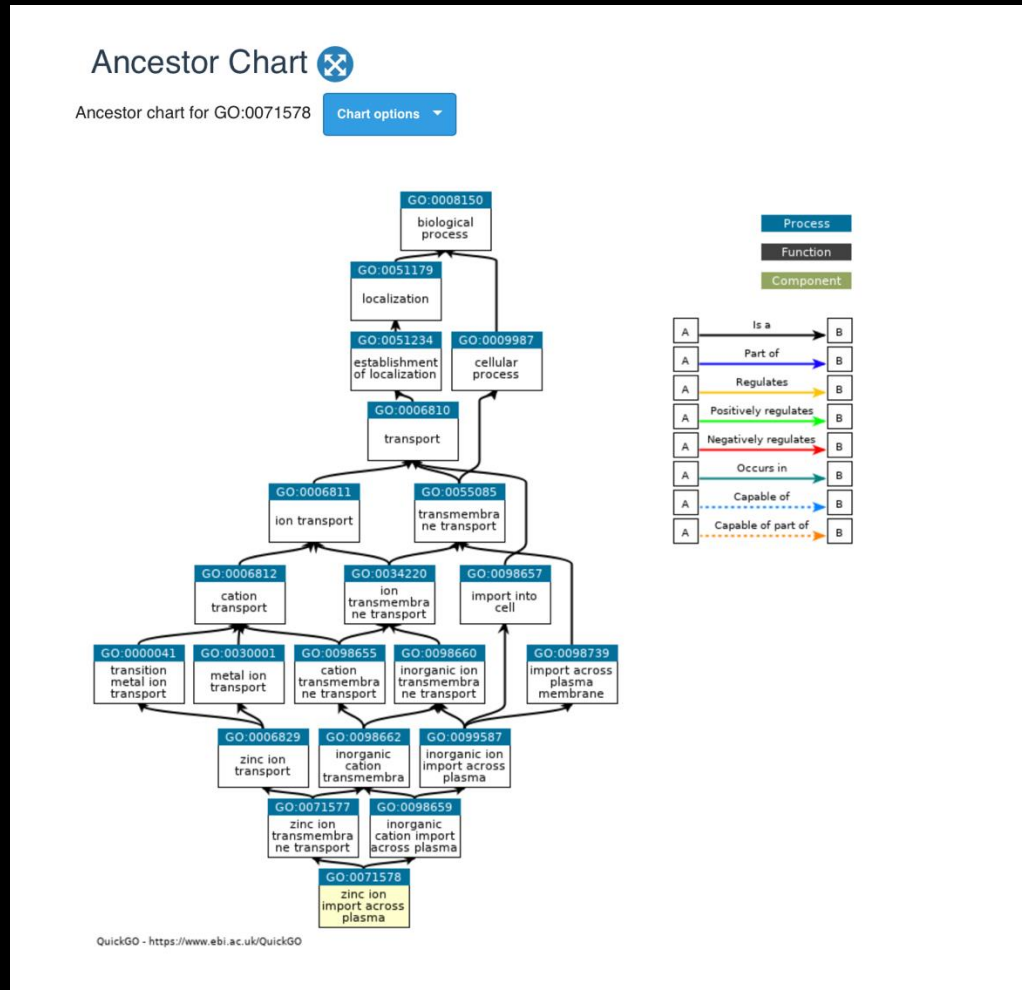
The screenshot displays the 'Inferred Tree View' of Gene Ontology (GO) terms. The interface includes a navigation bar with five tabs: 'Annotations', 'Graph Views', 'Inferred Tree View' (which is selected), 'Neighborhood', and 'Mappings'. The main content area shows a hierarchical tree structure of GO terms. The root term is 'GO:0008150 biological_process'. It has several children, including 'GO:0051179 localization' and 'GO:0006810 transport'. The 'transport' term is further expanded, showing a list of specific transport processes such as 'cellular process', 'establishment of localization', 'ion transport', 'transmembrane transport', 'cation transport', 'ion transmembrane transport', 'cation transmembrane transport', 'import across plasma membrane', 'import into cell', 'inorganic ion transmembrane transport', 'metal ion transport', 'transition metal ion transport', 'inorganic cation transmembrane transport', 'inorganic ion import across plasma membrane', 'zinc ion transport', 'inorganic cation import across plasma membrane', and 'zinc ion transmembrane transport'. The 'zinc ion transmembrane transport' term is expanded, revealing three sub-terms: 'GO:0071578 zinc ion import across plasma membrane' (which is highlighted in bold), 'GO:0071584 negative regulation of zinc ion transmembrane import', and 'GO:0071581 regulation of zinc ion transmembrane import'.

Annotations Graph Views Inferred Tree View Neighborhood Mappings

- GO:0008150 biological_process
 - GO:0051179 localization
 - GO:0006810 transport
 - GO:0009987 cellular process
 - GO:0051234 establishment of localization
 - GO:0006811 ion transport
 - GO:0055085 transmembrane transport
 - GO:0006812 cation transport
 - GO:0034220 ion transmembrane transport
 - GO:0098655 cation transmembrane transport
 - GO:0098739 import across plasma membrane
 - GO:0098657 import into cell
 - GO:0098660 inorganic ion transmembrane transport
 - GO:0030001 metal ion transport
 - GO:0000041 transition metal ion transport
 - GO:0098662 inorganic cation transmembrane transport
 - GO:0099587 inorganic ion import across plasma membrane
 - GO:0006829 zinc ion transport
 - GO:0098659 inorganic cation import across plasma membrane
 - GO:0071577 zinc ion transmembrane transport
 - GO:0071578 zinc ion import across plasma membrane**
 - GO:0071584 negative regulation of zinc ion transmembrane import
 - GO:0071581 regulation of zinc ion transmembrane import

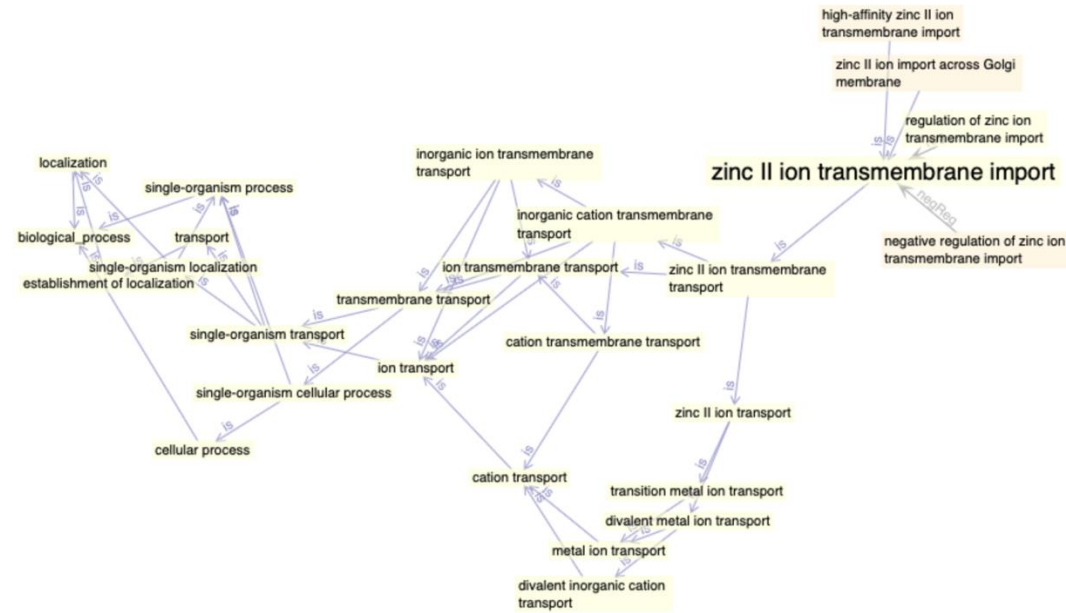
Gene Ontologies

An option from “Graph Views”



Gene Ontologies

Another option from “Graph Views”



Gene Ontologies – sources and evidence

Total annotations: 258; showing: 1-10
Results count

«First» «Prev» «Next» «Last» [Download \(up to 100000\)](#)

Gene/product	Gene/product name	Annotation qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence with	PANTHER family	Type	Isoform	Reference	Date
☐ BRCA1	Breast cancer type 1 susceptibility protein		ubiquitin ligase complex		UniProt	Homo sapiens	NAS		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:14976165	20050603
☐ BRCA1	Breast cancer type 1 susceptibility protein		double-strand break repair via homologous recombination		HGNC-UCL	Homo sapiens	IDA		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:17349954	20070426
☐ BRCA1	Breast cancer type 1 susceptibility protein		DNA double-strand break processing		Reactome	Homo sapiens	TAS		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		Reactome:R-HSA-5693608	20181122
☐ BRCA1	Breast cancer type 1 susceptibility protein		lateral element		MGI	Homo sapiens	IDA		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:9774970	20160830
☐ BRCA1	Breast cancer type 1 susceptibility protein		transcription regulatory region sequence-specific DNA binding		BHF-UCL	Homo sapiens	IDA		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:20820192	20110418
☐ BRCA1	Breast cancer type 1 susceptibility protein		DNA binding		PINC	Homo sapiens	TAS		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:9662397	20030904
☐ BRCA1	Breast cancer type 1 susceptibility protein		damaged DNA binding		Ensembl	Homo sapiens	IEA	UniProtKB:P48754 ensembl:ENSMUSP00000017290	breast cancer type 1 susceptibility protein brca1 pthr13763	protein		GO_REF:0000107	20201128
☐ BRCA1	Breast cancer type 1 susceptibility protein		transcription coactivator activity		UniProt	Homo sapiens	IMP		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:9662397	20170706
☐ BRCA1	Breast cancer type 1 susceptibility protein		RNA binding		MGI	Homo sapiens	IDA		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:12419249	20091111
☐ BRCA1	Breast cancer type 1 susceptibility protein		ubiquitin-protein transferase activity		UniProt	Homo sapiens	IDA		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:12890688	20100602

Experimental Evidence evidence codes:

Inferred from Experiment (EXP)

Inferred from Direct Assay (IDA)

Inferred from Physical Interaction (IPI)

Inferred from Mutant Phenotype (IMP)

Inferred from Genetic Interaction (IGI)

Inferred from Expression Pattern (IEP)

Computational Analysis evidence codes:

- Inferred from Sequence or structural Similarity (ISS)
- Inferred from Sequence Orthology (ISO)
- Inferred from Sequence Alignment (ISA)
- Inferred from Sequence Model (ISM)
- Inferred from Genomic Context (IGC)
- Inferred from Biological aspect of Ancestor (IBA)
- Inferred from Biological aspect of Descendant (IBD)
- Inferred from Key Residues (IKR)
- Inferred from Rapid Divergence (IRD)
- Inferred from Reviewed Computational Analysis (RCA)

Author Statement evidence codes:

Traceable Author Statement (TAS)

Non-traceable Author Statement (NAS)

Curatorial Statement codes:

Inferred by Curator (IC)

No biological Data available (ND) evidence code

Automatically-Assigned evidence code:

Inferred from Electronic Annotation (IEA)

PANTHER

<http://pantherdb.org>

ABOUT PANTHER

The PANTHER (Protein **AN**alysis **TH**rough **E**volutionary **R**elationships) Classification System was designed to classify proteins (and their genes) in order to facilitate high-throughput analysis. The core of PANTHER is a comprehensive, annotated "library" of gene family phylogenetic trees. All nodes in the tree have persistent identifiers that are maintained between versions of PANTHER, providing a stable substrate for annotations of protein properties like subfamily and function. Each phylogenetic tree is used to annotate each protein member of the family by its:

1. Family and Protein Class (supergrouping of protein families)
2. Subfamily (subgroup within the family phylogenetic tree)
3. Orthologs (genes in other organisms that derive from the same gene in the MRCA)
4. Paralogs (genes in the same organism that are related by gene duplication)
5. Function (using GO terms annotated on the trees by the [GO Phylogenetic Annotation Project](#))
6. Pathways (curated by PANTHER and by [Reactome](#))

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 PANTHER
Classification System

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POWERED

ABOUT PANTHER

The PANTHER (Protein **AN**alysis **TH**rough **E**volutionary **R**elationships) Classification System was designed to classify proteins (and their genes) in order to facilitate high-throughput analysis. Proteins have been classified according to:

- Family and subfamily: families are groups of evolutionarily related proteins; subfamilies are related proteins that also have the same function
- Molecular function: the function of the protein by itself or with directly interacting proteins at a biochemical level, e.g. a protein kinase
- Biological process: the function of the protein in the context of a larger network of proteins that interact to accomplish a process at the level of the cell or organism, e.g. mitosis.
- Pathway: similar to biological process, but a pathway also explicitly specifies the relationships between the interacting molecules.

The PANTHER Classifications are the result of human curation as well as sophisticated bioinformatics algorithms. Details of the methods can be found in ([Mi et al. NAR 2013](#); [Thomas et al., Genome Research 2003](#)).

For more information about the data on PANTHER and the relationships between data types, see the ["PANTHER Data Overview" figure](#).

PANTHER is part of the [Gene Ontology Phylogenetic Annotation Project](#).

PANTHER is supported by research grants from the National Human Genome Research Institute and the National Science Foundation, and maintained by the [Thomas lab at the University of Southern California](#).

Version 16.0 (released 2020-12-01) contains 15635 protein families, divided into 124388 functionally distinct protein subfamilies.

[PANTHER license and terms of use](#)

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Gene List Analysis

Browse

Sequence Search

cSNP Scoring

Keyword Search

Please refer to our article in [Nature Protocols](#) for detailed instructions on how to use this page.

Help Tips

Steps:

- 1. Select list and list type to analyze
- 2. Select Organism
- 3. Select operation

[Using enhancer data](#)

1. Enter ids and or select file for batch upload. Else enter ids or select file or list from workspace for comparing to a reference list.

Enter IDs:

Supported IDs

separate IDs by a space or comma

Upload IDs:

File format

Please [login](#) to be able to select lists from your workspace.

Select List Type:

☒ ID List

☐ Previously exported text search results

☐ Workspace list

☐ PANTHER Generic Mapping

☐ ID's from Reference Proteome Genome

Organism for id list

☐ VCF File

Flanking region

[Search Enhancer Data](#)

2. Select organism.

3. Select Analysis.

☒ Functional classification viewed in gene list

☐ Functional classification viewed in graphic charts

☐ Bar chart

☐ Pie chart

☐ Statistical overrepresentation test

☐ Statistical enrichment test

submit

GO enrichment analysis (PANTHER)

Enter a list of genes and click Launch >

The screenshot displays the Gene Ontology (GO) website. The header includes the GO logo, navigation links (About, Ontology, Annotations, Downloads, Help), and social media icons. A red banner at the top right contains a warning icon and the text "COVID-19 pandemic: click here to get GO data on SARS-CoV-2". Below the banner, the main heading "THE GENE ONTOLOGY RESOURCE" is followed by a paragraph about the GO Consortium's mission. A search bar is present with the placeholder text "Search GO term or Gene Product in AmiGO ...". To the right, the "GO Enrichment Analysis" tool is highlighted with a red rounded rectangle. This tool includes a text input field containing a list of gene symbols: APOH, APP, CND2, COL3A1, COL5A2, CXCL6, and EGF. Below the input field is a dropdown menu set to "biological process". At the bottom of the tool interface is a dropdown menu set to "Homo sapiens" and two buttons: "Examples" and "Launch >". A small text note at the bottom of the tool says "Gene set example: genes up-regulated during formation of blood vessels/angiogenesis (source: msigdb)".

GENE ONTOLOGY
Unifying Biology

About Ontology Annotations Downloads Help

COVID-19 pandemic: click here to get GO data on SARS-CoV-2

Current release 2021-02-01: 44,085 GO terms | 7,931,218 annotations
1,564,454 gene products | 4,743 species (see statistics)

THE GENE ONTOLOGY RESOURCE

The mission of the GO Consortium is to develop a comprehensive, **computational model of biological systems**, ranging from the molecular to the organism level, across the multiplicity of species in the tree of life.

The Gene Ontology (GO) knowledgebase is the world's largest source of information on the functions of genes. This knowledge is both human-readable and machine-readable, and is a foundation for computational analysis of large-scale molecular biology and genetics experiments in biomedical research.

Search GO term or Gene Product in AmiGO ...

Any ● Ontology ● Gene Product

GO Enrichment Analysis ?

Powered by PANTHER

APOH
APP
CND2
COL3A1
COL5A2
CXCL6
EGF

biological process

Homo sapiens Examples Launch >

Gene set example: genes up-regulated during formation of blood vessels/angiogenesis (source: msigdb)

GO enrichment analysis (PANTHER)

PANTHER results page. Notice you can select different gene sets/ontology sets and different statistical options.

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NewEnhancer-Gene MapPANTHER16.0 Released.

Analysis Summary: Please report in publication ?

Analysis Type: PANTHER Overrepresentation Test (Released 20210224)

Annotation Version and Release Date: GO Ontology database DOI: 10.5281/zenodo.4495804 Released 2021-02-01

Analyzed List:upload_1 (Homo sapiens)

Change

Reference List:Homo sapiens (all genes in database)

Change

Annotation Data Set:

GO biological process complete

GO molecular function complete

GO cellular component complete

PANTHER Pathways

PANTHER GO-Slim Molecular Function

PANTHER GO-Slim Biological Process

PANTHER GO-Slim Cellular Component

PANTHER Protein Class

Reactome pathways

?

Test Type:☒ Fisher's Exact

Correction:☒ Calculate False Discovery Rate☐ Use the Bonferroni correction for multiple testing ?☐ No correction

Results ?

Reference list

upload_1

Uniquely Mapped IDs:20595 out of 2059536 out of 39

Unmapped IDs:00

Multiple mapping information:03

ExportTableXML with user input idsJSON with user input ids

Displaying only results for FDR P < 0.05, [click here to display all results](#)

	Homo sapiens (REF)	upload_1 (Hierarchy, NEW! ?)					
	#	#	expected	Fold Enrichment	+/-	raw P value	FDR
GO biological process complete							
VEGF-activated neuropilin signaling pathway	2	2	.00	> 100	+	2.08E-05	2.81E-03
↳vascular endothelial growth factor signaling pathway	23	3	.04	68.88	+	1.57E-05	2.20E-03
↳transmembrane receptor protein tyrosine kinase signaling pathway	519	8	.98	8.14	+	5.26E-06	9.53E-04
↳enzyme linked receptor protein signaling pathway	726	11	1.37	8.00	+	7.53E-08	3.83E-05

Analysis Summary: Please report in publication ?

Analysis Type: PANTHER Overrepresentation Test (Released 20210224)

Annotation Version and Release Date: GO Ontology database DOI: 10.5281/zenodo.4495804 Released 2021-02-01

Analyzed List:upload_1 (Homo sapiens)

Change

Reference List:Homo sapiens (all genes in database)

Change

Annotation Data Set:GO biological process complete

?

Test Type:☒ Fisher's Exact☐ Binomial

Correction:☒ Calculate False Discovery Rate☐ Use the Bonferroni correction for multiple testing ?☐ No correction

GO enrichment analysis (PANTHER)

PANTHER results

- Test each annotation for enrichment
- Grouped by narrowest annotation category
- Sorted within group from most specific to most general

	Homo sapiens (REF)	#	# expected	Fold Enrichment	+/-	raw P-value	FDR
GO biological process complete		2	.00	> 100	+	2.08E-05	2.81E-03
VEGF-activated neuropilin signaling pathway		23	.04	68.88	+	1.57E-05	2.20E-03
↳vascular endothelial growth factor signaling pathway		519	.98	8.14	+	5.26E-06	9.53E-04
↳transmembrane receptor protein tyrosine kinase signaling pathway		726	1.37	8.00	+	7.53E-08	3.83E-05
↳enzyme linked receptor signaling pathway		2486	4.71	4.04	+	2.29E-08	1.44E-05
↳cell surface receptor signaling pathway		5138	23	9.73	+	6.76E-06	1.14E-03
↳signal transduction		5485	23	10.39	+	2.70E-05	3.31E-03
↳signaling		15487	39	29.33	+	2.07E-05	2.81E-03
↳cellular process		5572	24	10.55	+	1.12E-05	1.69E-03
↳cell communication		6770	26	12.82	+	1.99E-05	2.73E-03
↳cellular response to stimulus		8496	35	16.09	+	3.82E-10	1.21E-06
↳response to stimulus		11302	33	21.40	+	1.39E-04	1.21E-02
↳regulation of cellular process		11856	33	22.45	+	4.95E-04	3.15E-02
↳regulation of biological process		12571	34	23.81	+	4.61E-04	3.04E-02
↳biological regulation		44	3	.08	+	9.56E-05	9.03E-03
↳cellular response to vascular endothelial growth factor stimulus		503	10	.95	+	2.75E-08	1.67E-05
↳cellular response to growth factor stimulus		2331	18	4.41	+	5.67E-08	3.08E-05
↳cellular response to organic substance		2898	19	5.49	+	2.71E-07	9.29E-05
↳cellular response to chemical stimulus		4369	25	8.27	+	9.54E-09	7.53E-06
↳response to chemical		2995	22	5.67	+	1.61E-09	2.31E-06
↳response to organic substance		532	11	1.01	+	3.26E-09	3.67E-06
↳response to growth factor		4	2	.01	+	5.20E-05	5.50E-03
↳neuropilin signaling pathway		2	2	.00	+	2.08E-05	2.79E-03
positive regulation of retinal ganglion cell axon guidance		3	2	.01	+	3.47E-05	4.05E-03
↳positive regulation of axon guidance		14	2	.03	+	4.11E-04	2.78E-02
↳regulation of axon guidance		1026	14	1.94	+	2.89E-09	3.51E-06
↳regulation of cellular component movement		2734	18	5.18	+	6.39E-07	1.87E-04
↳regulation of localization		222	5	.42	+	6.54E-05	6.53E-03
↳regulation of chemotaxis		990	14	1.87	+	1.84E-09	2.42E-06
↳regulation of locomotion		1182	12	2.24	+	1.21E-06	2.81E-04
↳regulation of response to external stimulus		4240	26	8.03	+	6.93E-10	1.37E-06
↳regulation of response to stimulus		556	9	1.05	+	8.27E-07	2.14E-04
↳positive regulation of cellular component movement		5656	21	10.71	+	5.11E-04	3.22E-02
↳positive regulation of cellular process		6244	26	11.82	+	3.91E-06	7.82E-04
↳positive regulation of biological process		5	2	.01	+	7.27E-05	7.12E-03
↳regulation of retinal ganglion cell axon guidance		6	2	.01	+	9.68E-05	9.09E-03
motor neuron migration		965	13	1.83	+	1.45E-08	9.94E-06
↳cell migration		1103	14	2.09	+	7.22E-09	6.33E-06
↳cell motility		1591	16	3.01	+	1.14E-08	8.20E-06
↳movement of cell or subcellular component		1103	14	2.09	+	7.22E-09	6.70E-06
↳localization of cell		5807	26	11.00	+	7.01E-07	1.91E-04
↳localization		1339	16	2.54	+	9.76E-10	1.71E-06
↳locomotion		1249	11	2.37	+	1.44E-05	2.07E-03
↳generation of neurons		1369	12	2.59	+	5.53E-06	9.79E-04
↳neurogenesis		3509	18	6.64	+	2.38E-05	3.02E-03
↳cell differentiation		3562	18	6.75	+	2.93E-05	3.50E-03
↳cellular developmental process		5729	26	10.85	+	5.26E-07	1.57E-04
↳developmental process		2189	14	4.15	+	2.72E-05	3.30E-03
↳nervous system development		4279	24	8.10	+	3.94E-08	2.22E-05
↳system development		5266	26	9.97	+	8.61E-08	4.24E-05
↳anatomical structure development		4877	26	9.24	+	1.60E-08	1.05E-05
↳multicellular organism development		6909	32	13.08	+	6.78E-10	1.53E-06
↳multicellular organismal process							

KEGG

<https://www.genome.jp/kegg/>

KEGG is a set of related databases.

KEGG PATHWAY is the most important part for our purposes.



KEGG

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KEGG: Kyoto Encyclopedia of Genes and Genomes

KEGG is a database resource for understanding high-level functions and utilities of the biological system, such as the cell, the organism and the ecosystem, from molecular-level information, especially large-scale molecular datasets generated by genome sequencing and other high-throughput experimental technologies. See [Release notes](#) (April 1, 2021) for new and updated features.

New article [KEGG: integrating viruses and cellular organisms](#)

Main entry point to the KEGG web service

KEGG2 [KEGG Table of Contents](#) [[Update notes](#) | [Release history](#)]

Data-oriented entry points

KEGG PATHWAY [KEGG pathway maps](#)

KEGG BRITE [BRITE hierarchies and tables](#)

KEGG MODULE [KEGG modules](#)

KEGG ORTHOLOGY [KO functional orthologs](#) [[Annotation](#)]

KEGG GENOME [Genomes](#) [[Pathogen](#) | [Virus](#) | [Plant](#)]

KEGG GENES [Genes and proteins](#) [[SeqData](#)]

KEGG COMPOUND [Small molecules](#)

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KEGG REACTION [Biochemical reactions](#) [[RModule](#)]

KEGG ENZYME [Enzyme nomenclature](#)

KEGG NETWORK [Disease-related network variations](#)

KEGG DISEASE [Human diseases](#)

KEGG DRUG [Drugs](#) [[New drug approvals](#)]

KEGG MEDICUS [Health information resource](#) [[Drug labels search](#)]

Organism-specific entry points

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Analysis tools

KEGG Mapper [KEGG PATHWAY/BRITE/MODULE mapping tools](#)

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GhostKOALA [GHOSTX-based KO annotation and KEGG mapping](#)

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KEGG PATHWAY

A collection of manually drawn pathway maps.

Pathways fall into 7 categories.



KEGG PATHWAY Database

Wiring diagrams of molecular interactions, reactions and relations

[Menu](#) [PATHWAY](#) [BRITE](#) [MODULE](#) [KO](#) [GENES](#) [LIGAND](#) [NETWORK](#) [DISEASE](#) [DRUG](#) [DBGET](#)

Select prefix

map

Organism

Enter keywords

Go

[Help](#)

[\[New pathway maps | Update history \]](#)

Pathway Maps

KEGG PATHWAY is a collection of manually drawn [pathway maps](#) representing our knowledge of the molecular interaction, reaction and relation networks for:

1. Metabolism

[Global/overview](#) [Carbohydrate](#) [Energy](#) [Lipid](#) [Nucleotide](#) [Amino acid](#) [Other amino](#) [Glycan](#)
[Cofactor/vitamin](#) [Terpenoid/PK](#) [Other secondary metabolite](#) [Xenobiotics](#) [Chemical structure](#)

2. Genetic Information Processing

3. Environmental Information Processing

4. Cellular Processes

5. Organismal Systems

6. Human Diseases

7. Drug Development

KEGG PATHWAY is the reference database for pathway mapping in [KEGG Mapper](#).

Pathway Identifiers

Each pathway map is identified by the combination of 2-4 letter prefix code and 5 digit number (see [KEGG Identifier](#)). The prefix has the following meaning:

map	manually drawn reference pathway
ko	reference pathway highlighting KOs
ec	reference metabolic pathway highlighting EC numbers
rn	reference metabolic pathway highlighting reactions
<org>	organism-specific pathway generated by converting KOs to gene identifiers

and the numbers starting with the following:

011	global map (lines linked to KOs)
012	overview map (lines linked to KOs)
010	chemical structure map (no KO expansion)
07	drug structure map (no KO expansion)
other	regular map (boxes linked to KOs)

are used for different types of maps.

KEGG PATHWAY is integrated with [MODULE](#) and [NETWORK](#) databases as indicated below.

M	- module
R	- reaction module
N	- network

KEGG PATHWAY

A few of the KEGG pathways.

Let's select "Metabolic pathways"
(the entire category).

1. Metabolism

1.0 Global and overview maps

- 01100 M **Metabolic pathways**
- 01110 M Biosynthesis of secondary metabolites
- 01120 M Microbial metabolism in diverse environments
- 01200 M R Carbon metabolism
- 01210 M R 2-Oxocarboxylic acid metabolism
- 01212 M R Fatty acid metabolism
- 01230 M R Biosynthesis of amino acids
- 01240 M R Biosynthesis of cofactors *New!*
- 01220 M R Degradation of aromatic compounds

1.1 Carbohydrate metabolism

- 00010 M N Glycolysis / Gluconeogenesis
- 00020 M Citrate cycle (TCA cycle)
- 00030 M Pentose phosphate pathway
- 00040 M Pentose and glucuronate interconversions
- 00051 M Fructose and mannose metabolism
- 00052 M N Galactose metabolism
- 00053 M Ascorbate and aldarate metabolism
- 00500 M N Starch and sucrose metabolism
- 00520 M N Amino sugar and nucleotide sugar metabolism
- 00620 M Pyruvate metabolism
- 00630 M Glyoxylate and dicarboxylate metabolism
- 00640 M Propanoate metabolism
- 00650 M Butanoate metabolism
- 00660 M C5-Branched dibasic acid metabolism
- 00562 M Inositol phosphate metabolism

1.2 Energy metabolism

- 00190 M Oxidative phosphorylation
- 00195 M Photosynthesis
- 00196 Photosynthesis - antenna proteins
- 00710 M Carbon fixation in photosynthetic organisms
- 00720 M Carbon fixation pathways in prokaryotes
- 00680 M Methane metabolism
- 00910 M Nitrogen metabolism
- 00920 M Sulfur metabolism

1.3 Lipid metabolism

- 00061 M Fatty acid biosynthesis
- 00062 M Fatty acid elongation
- 00071 M N Fatty acid degradation
- 00072 M Synthesis and degradation of ketone bodies
- 00073 Cutin, suberine and wax biosynthesis
- 00100 M Steroid biosynthesis
- 00120 M N Primary bile acid biosynthesis
- 00121 Secondary bile acid biosynthesis
- 00140 M N Steroid hormone biosynthesis
- 00561 M Glycerolipid metabolism
- 00564 M Glycerophospholipid metabolism
- 00565 M Ether lipid metabolism
- 00600 M N Sphingolipid metabolism
- 00590 Arachidonic acid metabolism
- 00591 Linoleic acid metabolism
- 00592 M alpha-Linolenic acid metabolism
- 01040 M N Biosynthesis of unsaturated fatty acids

1.4 Nucleotide metabolism

- 00230 M N Purine metabolism
- 00240 M Pyrimidine metabolism

1.5 Amino acid metabolism

- 00250 M Alanine, aspartate and glutamate metabolism
- 00260 M Glycine, serine and threonine metabolism
- 00270 M Cysteine and methionine metabolism
- 00280 M N Valine, leucine and isoleucine degradation
- 00290 M Valine, leucine and isoleucine biosynthesis
- 00300 M Lysine biosynthesis
- 00310 M Lysine degradation
- 00220 M N Arginine biosynthesis
- 00330 M Arginine and proline metabolism

Change pathway type

▼ Option

Scale: 30%

Link: Normal

▼ Search

Go

▼ User data

▼ Module

Pathway modules

Carbohydrate metabolism

Central carbohydrate metabolism

M00001 Glycolysis (Embden-Meyerhof pathway)

M00002 Glycolysis, core metabolism

M00003 Gluconeogenesis

M00307 Pyruvate oxidation

M00009 Citrate cycle (TCA cycle)

M00010 Citrate cycle, first half

M00011 Citrate cycle, second half

M00004 Pentose phosphate pathway

M00006 Pentose phosphate pathway

M00007 Pentose phosphate pathway

M00580 Pentose phosphate pathway

M00005 PRPP biosynthesis

M00008 Entner-Doudoroff pathway

M00308 Semi-phosphorylated sugar metabolism

M00633 Semi-phosphorylated sugar metabolism

M00309 Non-phosphorylated sugar metabolism

Other carbohydrate metabolism

M00014 Glucuronate pathway

M00630 D-Galacturonate degradation

M00631 D-Galacturonate degradation

M00061 D-Glucuronate degradation

M00081 Pectin degradation

M00632 Galactose degradation

M00552 D-galactonate degradation

M00129 Ascorbate biosynthesis

M00114 Ascorbate biosynthesis

M00550 Ascorbate degradation

M00854 Glycogen biosynthesis

M00855 Glycogen degradation

M00565 Trehalose biosynthesis

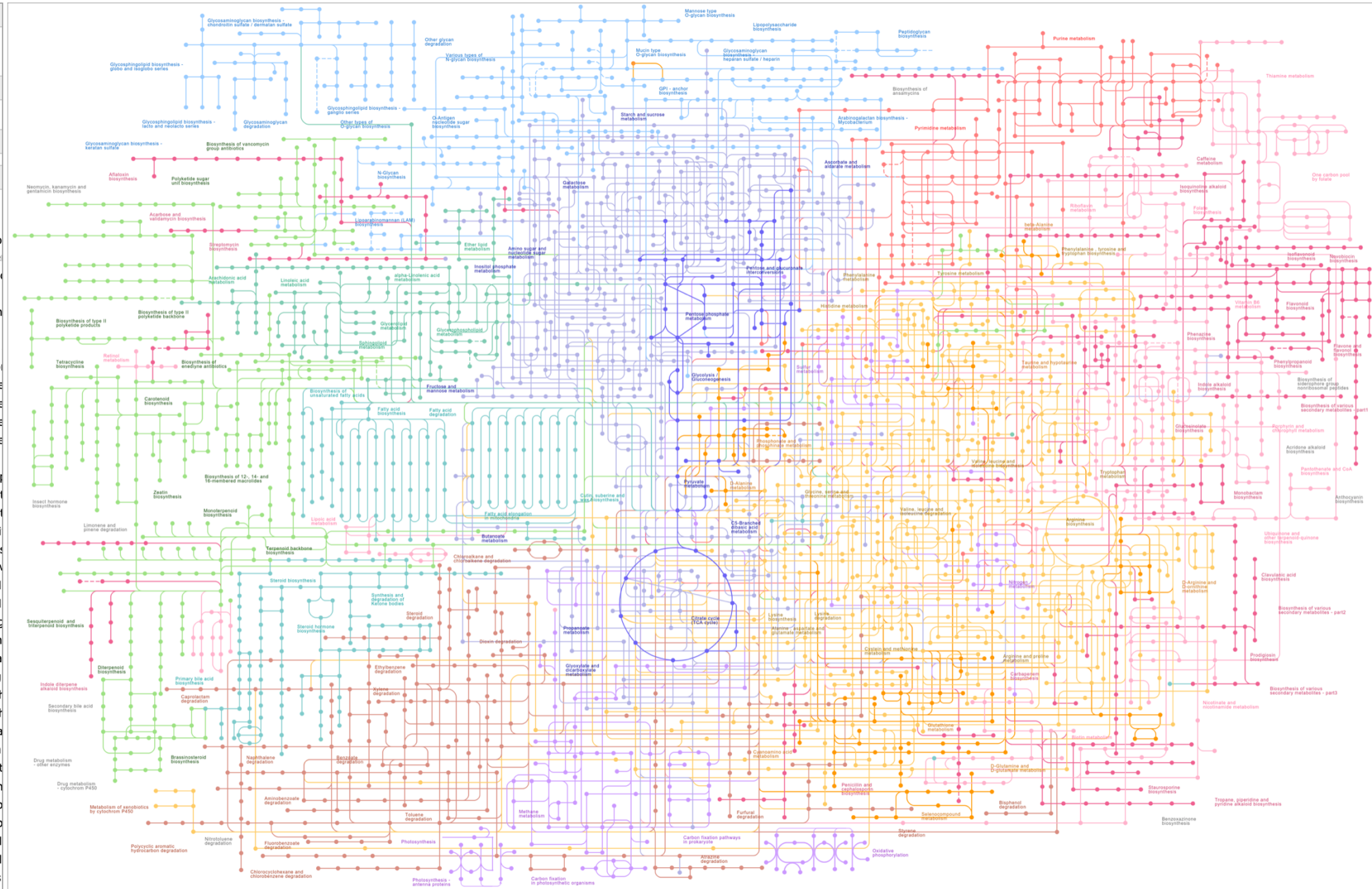
M00549 Nucleotide sugar biosynthesis

M00554 Nucleotide sugar biosynthesis

M00892 UDP-N-acetyl-D-glucosamine biosynthesis

M00909 UDP-N-acetyl-D-glucosamine biosynthesis

M00761 Undecaprenylphosphate biosynthesis



Change pathway type

▼ Option

Scale: 30%

Link:

▼ Search

▼ User data

▼ Module

☐ Pathway modules

☐ Carbohydrate metabolism

☐ Central carbohydrate metabo

☐ M00001 Glycolysis (Embd

☐ M00002 Glycolysis, core m

☐ M00003 Gluconeogenesis

☐ M00307 Pyruvate oxidation

☐ M00009 Citrate cycle (TCA

☐ M00010 Citrate cycle, first

☐ M00011 Citrate cycle, seco

☐ M00004 Pentose phosphate

☐ M00006 Pentose phosphate

☐ M00007 Pentose phosphate

☐ M00580 Pentose phosphate

☐ M00005 PRPP biosynthesis

☐ M00008 Entner-Doudoroff p

☐ M00308 Semi-phosphorylat

☐ M00633 Semi-phosphorylat

☐ M00309 Non-phosphorylati

☐ Other carbohydrate metaboli

☐ M00014 Glucuronate pathw

☐ M00630 D-Galacturonate d

☐ M00631 D-Galacturonate d

☐ M00061 D-Glucuronate deg

☐ M00081 Pectin degradation

☐ M00632 Galactose degrada

☐ M00552 D-galactonate deg

☐ M00129 Ascorbate biosynt

☐ M00114 Ascorbate biosynt

☐ M00550 Ascorbate degrada

☐ M00854 Glycogen biosynt

☐ M00855 Glycogen degradat

☐ M00565 Trehalose biosynt

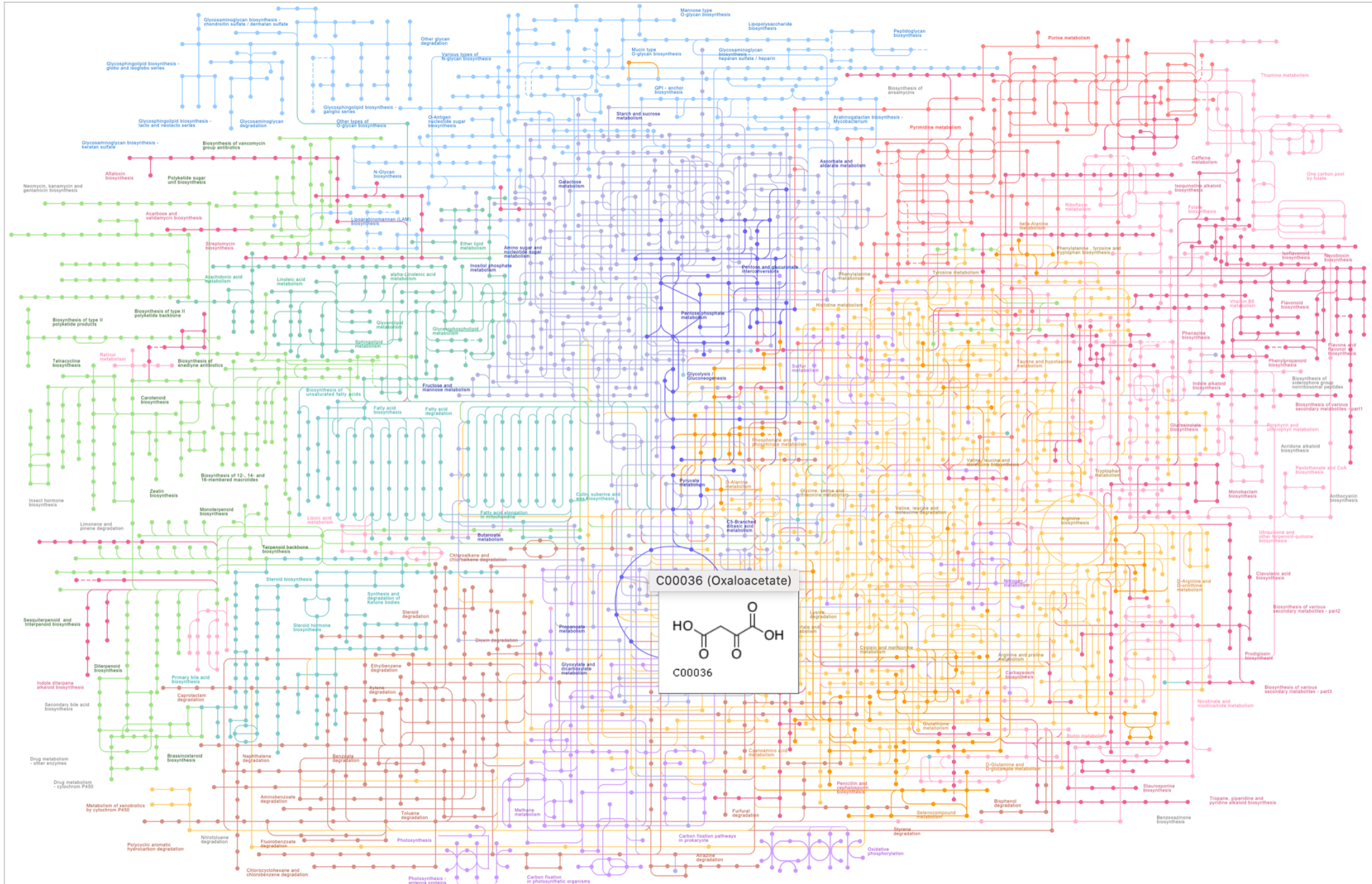
☐ M00549 Nucleotide sugar b

☐ M00554 Nucleotide sugar b

☐ M00892 UDP-N-acetyl-D-gl

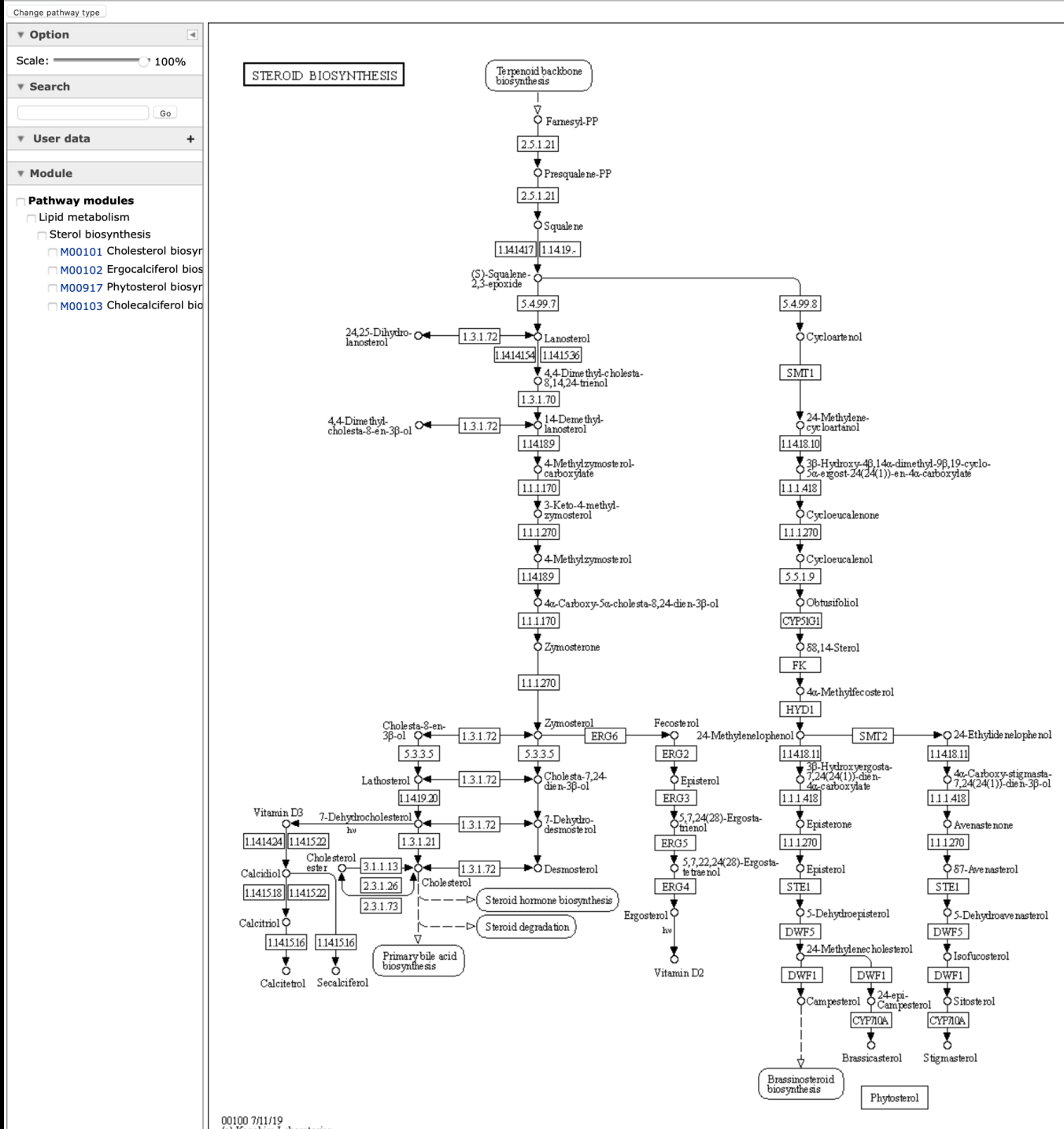
☐ M00909 UDP-N-acetyl-D-gl

☐ M00761 Undecaprenylphos



KEGG

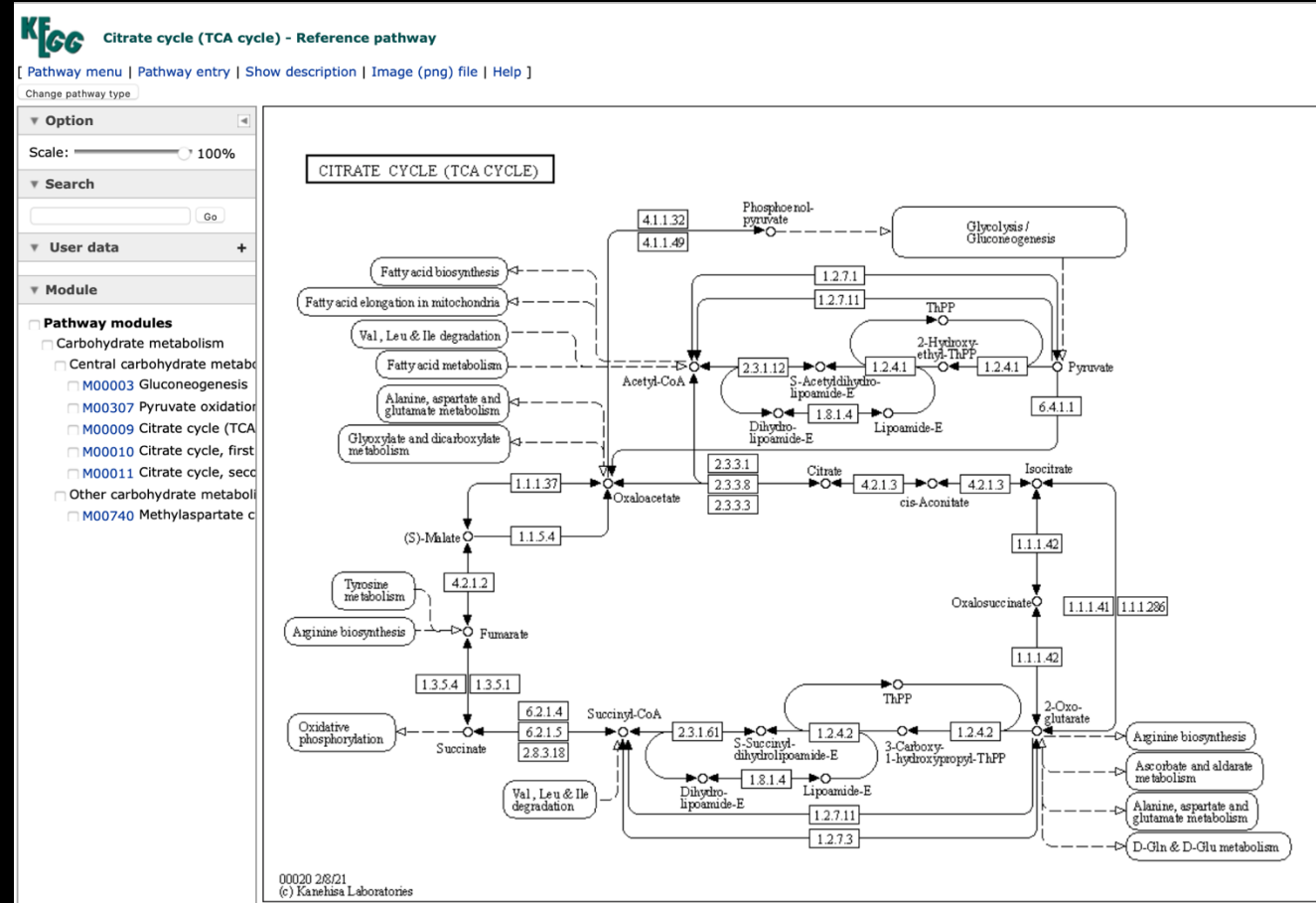
Selecting the “steroid biosynthesis” pathway



KEGG

Selecting TCA cycle

Click on a component, e.g., succinate



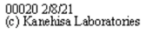
Entry	C00042	Compound
Name	Succinate; Succinic acid; Butanedionic acid; Ethylenesuccinic acid	
Formula	C4H6O4	
Exact mass	118.0266	
Mol weight	118.088	
Structure	 C00042 Mol file KCF file DB search	
Reaction	R00402 R00405 R00406 R00407 R00408 R00409 R00410 R00411 R00412 R00432 R00479 R00499 R00713 R00714 R00727 R00999 R01252 R01288 R01412 R01559 R01780 R01867 R01879 R02084 R02160 R02164 R02397 R02407 R02444 R02445 R02486 R02508 R02603 R02734 R02990 R03008 R03009 R03126 R03154 R03218 R03219 R03260 R03376 R03451 R03590 R03640 R03698 R03737 R03806 R03807 R03809 R03812 R04073 R04276 R04946 R05036 R05037 R05039 R05097 R05126 R05127 R05228 R05229 R05301 R05302 R05320 R05419 R05466 R05468 R05469 R05493 R05588 R05722 R05723 R05857 R06322 R06323 R06335 R06336 R06337 » show all	
Pathway	map00020 Citrate cycle (TCA cycle) map00190 Oxidative phosphorylation map00250 Alanine, aspartate and glutamate metabolism map00310 Lysine degradation map00350 Tyrosine metabolism map00360 Phenylalanine metabolism map00361 Chlorocyclohexane and chlorobenzene degradation map00620 Pyruvate metabolism map00630 Glyoxylate and dicarboxylate metabolism map00640 Propanoate metabolism map00650 Butanoate metabolism map00720 Carbon fixation pathways in prokaryotes map00760 Nicotinate and nicotinamide metabolism map00920 Sulfur metabolism map01060 Biosynthesis of plant secondary metabolites map01061 Biosynthesis of phenylpropanoids map01062 Biosynthesis of terpenoids and steroids map01063 Biosynthesis of alkaloids derived from shikimate pathway map01064 Biosynthesis of alkaloids derived from ornithine, lysine and nicotinic acid map01065 Biosynthesis of alkaloids derived from histidine and purine map01066 Biosynthesis of alkaloids derived from terpenoid and polyketide map01070 Biosynthesis of plant hormones map01100 Metabolic pathways map01110 Biosynthesis of secondary metabolites map01120 Microbial metabolism in diverse environments map01200 Carbon metabolism map01220 Degradation of aromatic compounds map02020 Two-component system map04024 cAMP signaling pathway map04727 GABAergic synapse map04922 Glucagon signaling pathway map05230 Central carbon metabolism in cancer	
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate M00027 GABA (gamma-Aminobutyrate) shunt M00148 Succinate dehydrogenase (ubiquinone) M00149 Succinate dehydrogenase, prokaryotes M00150 Fumarate reductase, prokaryotes M00173 Reductive citrate cycle (Arnon-Buchanan cycle) M00374 Dicarboxylate-hydroxybutyrate cycle M00376 3-Hydroxypropionate bi-cycle M00620 Incomplete reductive citrate cycle, acetyl-CoA => oxoglutarate	

All links
Ontology (1) KEGG BRITE (1) Pathway (42) KEGG PATHWAY (32) KEGG MODULE (10) Network (1) KEGG NETWORK (1) Drug (9) KEGG ENVIRON (8) ChEMBL (1) Chemical substance (25) PubChem (1) ChEBI (1) 3DMET (1) HMDB (1) HSDB (1) KNApSACk (1) LIPIDMAPS (1) MASSBANK (16) NIKKAJI (1) PDB-CCD (1) Chemical reaction (279) KEGG ENZYME (104) KEGG REACTION (175) Gene (90353) KEGG GENES (90353) All databases (90710) Download RDF

succinate

Module	M00009 Citrate cycle (TCA cycle, Krebs cycle)			
	M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate			
	M00027 GABA (gamma-Aminobutyrate) shunt			
	M00148 Succinate dehydrogenase (ubiquinone)			
	M00149 Succinate dehydrogenase, prokaryotes			
	M00150 Fumarate reductase, prokaryotes			
	M00173 Reductive citrate cycle (Arnon-Buchanan cycle)			
	M00374 Dicarboxylate-hydroxybutyrate cycle			
	M00376 3-Hydroxypropionate bi-cycle			
	M00620 Incomplete reductive citrate cycle, acetyl-CoA => oxoglutarate			
Enzyme	1.2.1.16	1.2.1.24	1.2.1.79	1.3.1.1
	1.3.4.1	1.3.5.1	1.3.5.4	1.3.5.5
	1.3.99.-	1.14.11.1	1.14.11.2	1.14.11.3
	1.14.11.4	1.14.11.6	1.14.11.7	1.14.11.8
	1.14.11.9	1.14.11.10	1.14.11.11	1.14.11.12
	1.14.11.13	1.14.11.15	1.14.11.16	1.14.11.17
	1.14.11.18	1.14.11.20	1.14.11.21	1.14.11.22
	1.14.11.25	1.14.11.26	1.14.11.27	1.14.11.28
	1.14.11.31	1.14.11.32	1.14.11.35	1.14.11.36
	1.14.11.37	1.14.11.38	1.14.11.39	1.14.11.40
	» show all			
Brite	Compounds with biological roles [BR:br08001]			
	Organic acids Carboxylic acids Dicarboxylic acids C00042 Succinate BRITE hierarchy			
Other DBs	CAS: 110-15-6 PubChem: 3344 ChEBI: 15741 ChEMBL: CHEMBL576 LIPIDMAPS: LMFA01170043 KNApSACk: C00001205 PDB-CCD: SIN[PDBj] 3DMET: B00012 NIKKAJI: J2.879G			
KCF data	Show			

Go back to the map and choose
and enzyme (6.2.1.4)



KEGG

 ORTHOLOGY: K01899		Help
Entry	K01899 KO	
Name	LSC1	
Definition	succinyl-CoA synthetase alpha subunit [EC:6.2.1.4 6.2.1.5]	
Pathway	ko00020 Citrate cycle (TCA cycle) ko00640 Propanoate metabolism ko01100 Metabolic pathways ko01110 Biosynthesis of secondary metabolites ko01120 Microbial metabolism in diverse environments ko01200 Carbon metabolism	
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate	
Disease	H00469 Mitochondrial DNA depletion syndrome	
Brite	KEGG Orthology (KO) [BR:ko00001] 09100 Metabolism 09101 Carbohydrate metabolism 00020 Citrate cycle (TCA cycle) K01899 LSC1; succinyl-CoA synthetase alpha subunit 00640 Propanoate metabolism K01899 LSC1; succinyl-CoA synthetase alpha subunit Enzymes [BR:ko01000] 6. Ligases 6.2 Forming carbon-sulfur bonds 6.2.1 Acid-thiol ligases 6.2.1.4 succinate---CoA ligase (GDP-forming) K01899 LSC1; succinyl-CoA synthetase alpha subunit 6.2.1.5 succinate---CoA ligase (ADP-forming) K01899 LSC1; succinyl-CoA synthetase alpha subunit BRITE hierarchy	
Other DBs	RN: R00432 R00727 GO: 0004776 0004775	
Genes	HSA: 8802 (SUCLG1) PTR: 470417 (SUCLG1) PPS: 100985425 (SUCLG1) GGO: 101143056 (SUCLG1) PON: 100173831 (SUCLG1) NLE: 100579244 (SUCLG1) MCC: 693991 (SUCLG1) MCF: 101867049 (SUCLG1) CSAB: 103219966 (SUCLG1) RRO: 104665951 (SUCLG1) » show all Taxonomy UniProt	

All links

Ontology (4)
KEGG BRITE (2)
GO (2)
Pathway (14)
KEGG PATHWAY (12)
KEGG MODULE (2)
Disease (1)
KEGG DISEASE (1)
Chemical reaction (6)
KEGG ENZYME (2)
KEGG REACTION (2)
KEGG RCLASS (2)
Gene (10967)
KEGG GENES (718)
KEGG MGENES (468)
RefGene (9743)
OC (38)
[All databases \(10992\)](#)
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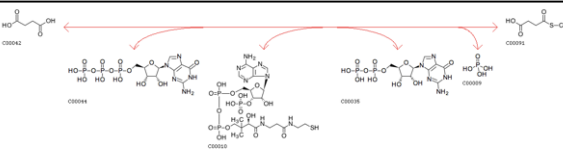
 ORTHOLOGY: K01900		Help
Entry	K01900 KO	
Name	LSC2	
Definition	succinyl-CoA synthetase beta subunit [EC:6.2.1.4 6.2.1.5]	
Pathway	ko00020 Citrate cycle (TCA cycle) ko00640 Propanoate metabolism ko01100 Metabolic pathways ko01110 Biosynthesis of secondary metabolites ko01120 Microbial metabolism in diverse environments ko01200 Carbon metabolism	
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate	
Disease	H00469 Mitochondrial DNA depletion syndrome	
Brite	KEGG Orthology (KO) [BR:ko00001] 09100 Metabolism 09101 Carbohydrate metabolism 00020 Citrate cycle (TCA cycle) K01900 LSC2; succinyl-CoA synthetase beta subunit 00640 Propanoate metabolism K01900 LSC2; succinyl-CoA synthetase beta subunit 09180 Brite Hierarchies 09183 Protein families: signaling and cellular processes 04147 Exosome K01900 LSC2; succinyl-CoA synthetase beta subunit Enzymes [BR:ko01000] 6. Ligases 6.2 Forming carbon-sulfur bonds 6.2.1 Acid-thiol ligases 6.2.1.4 succinate---CoA ligase (GDP-forming) K01900 LSC2; succinyl-CoA synthetase beta subunit 6.2.1.5 succinate---CoA ligase (ADP-forming) K01900 LSC2; succinyl-CoA synthetase beta subunit Exosome [BR:ko04147] Exosomal proteins Exosomal proteins of other body fluids (saliva and other) K01900 LSC2; succinyl-CoA synthetase beta subunit BRITE hierarchy	
Other DBs	RN: R00432 R00727 GO: 0004776 0004775	
Genes	HSA: 8801 (SUCLG2) 8803 (SUCLA2) PTR: 452711 (SUCLA2) 460492 (SUCLG2) PPS: 100971297 (SUCLA2) 100973485 (SUCLG2) GGO: 101127943 (SUCLG2) 101128782 (SUCLA2) PON: 100171525 (SUCLA2) 100443464 (SUCLG2) NLE: 100584620 (SUCLA2) 100595372 (SUCLG2) MCC: 697454 (SUCLG2) 704434 (SUCLA2) MCF: 101867314 (SUCLA2) 102144513 (SUCLG2) CSAB: 103214398 (SUCLA2) 103227863 (SUCLG2) RRO: 104660507 104661533 104662766 (SUCLA2) 104678622 » show all Taxonomy UniProt	

All links

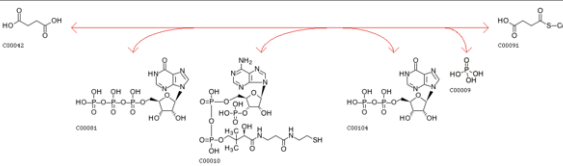
Ontology (5)
KEGG BRITE (3)
GO (2)
Pathway (14)
KEGG PATHWAY (12)
KEGG MODULE (2)
Disease (1)
KEGG DISEASE (1)
Chemical reaction (6)
KEGG ENZYME (2)
KEGG REACTION (2)
KEGG RCLASS (2)
Gene (12528)
KEGG GENES (908)
KEGG MGENES (551)
RefGene (11044)
OC (25)
[All databases \(12554\)](#)
[Download RDF](#)

	ENZYME: 6.2.1.4	Help
Entry	EC 6.2.1.4 Enzyme	
Name	succinate--CoA ligase (GDP-forming); succinyl-CoA synthetase (GDP-forming); succinyl coenzyme A synthetase (guanosine diphosphate-forming); succinate thiokinase (ambiguous); succinic thiokinase (ambiguous); succinyl coenzyme A synthetase (ambiguous); succinate-phosphorylating enzyme (ambiguous); P-enzyme; SCS (ambiguous); G-STK; succinyl coenzyme A synthetase (GDP-forming); succinyl CoA synthetase (ambiguous)	
Class	Ligases; Forming carbon-sulfur bonds; Acid-thiol ligases BRITE hierarchy	
Sysname	succinate:CoA ligase (GDP-forming)	
Reaction(IUBMB)	GTP + succinate + CoA = GDP + phosphate + succinyl-CoA [RN:R00432]	
Reaction(KEGG)	R00432; (other) R00727 R02405 R02406 Reaction	
Substrate	GTP [CPD:C00044]; succinate [CPD:C00042]; CoA [CPD:C00010]	
Product	GDP [CPD:C00035]; phosphate [CPD:C00009]; succinyl-CoA [CPD:C00091]	
Comment	Itaconate can act instead of succinate, and ITP instead of GTP.	
History	EC 6.2.1.4 created 1961	
Pathway	ec00020 Citrate cycle (TCA cycle) ec00640 Propanoate metabolism ec01100 Metabolic pathways ec01110 Biosynthesis of secondary metabolites ec01120 Microbial metabolism in diverse environments	
Orthology	K01899 succinyl-CoA synthetase alpha subunit K01900 succinyl-CoA synthetase beta subunit	
Genes	HSA: 8801(SUCLG2) 8802(SUCLG1) 8803(SUCLA2) PTR: 452711(SUCLA2) 460492(SUCLG2) 470417(SUCLG1) PPS: 100971297(SUCLA2) 100973485(SUCLG2) 100985425(SUCLG1) GGO: 101127943(SUCLG2) 101128782(SUCLA2) 101143056(SUCLG1) PON: 100171525(SUCLA2) 100173831(SUCLG1) 100443464(SUCLG2) NLE: 100579244(SUCLG1) 100584620(SUCLA2) 100595372(SUCLG2) MCC: 693991(SUCLG1) 697454(SUCLG2) 704434(SUCLA2) MCF: 101867049(SUCLG1) 101867314(SUCLA2) 102144513(SUCLG2) CSAB: 103214398(SUCLA2) 103219966(SUCLG1) 103227863(SUCLG2) RRO: 104660507 104661533 104662766(SUCLA2) 104665951(SUCLG1) 104678622 » show all Taxonomy	
Reference Authors Title Journal	1 Hager, L.P. Succinyl CoA synthetase. In: Boyer, P.D., Lardy, H. and Myrback, K. (Eds.), The Enzymes, 2nd ed., vol. 6, Academic Press, New York, 1962, p. 387-399.	
Reference Authors Title Journal	2 [PMID:13084656] KAUFMAN S, GILVARG C, CORI O, OCHOA S. Enzymatic oxidation of alpha-ketoglutarate and coupled phosphorylation. J Biol Chem 203:869-88 (1953)	
Reference Authors Title Journal	3 [PMID:13768680] MAZUMDER R, SANADI DR, RODWELL VW. Purification and properties of hog kidney succinic thiokinase. J Biol Chem 235:2546-50 (1960)	
Reference Authors Title Journal	4 [PMID:13181903] SANADI DR, GIBSON M, AYENGAR P. Guanosine triphosphate, the primary product of phosphorylation coupled to the breakdown of succinyl coenzyme A. Biochim Biophys Acta 14:434-6 (1954) DOI:10.1016/0006-3002(54)90205-X	
Other DBs	ExplorEnz - The Enzyme Database: 6.2.1.4 IUBMB Enzyme Nomenclature: 6.2.1.4 ExPASy - ENZYME nomenclature database: 6.2.1.4 BRENDA, the Enzyme Database: 6.2.1.4 CAS: 9014-36-2	

All links
Pathway (12) KEGG PATHWAY (10) KEGG MODULE (2) Chemical substance (10) KEGG COMPOUND (10) Chemical reaction (6) KEGG REACTION (4) KEGG RCLASS (2) Gene (23434) KEGG ORTHOLOGY (2) KEGG GENES (1626) KEGG MGENES (1019) RefGene (20787) Protein sequence (1749) UniProt (1613) SWISS-PROT (18) RefSeq(pep) (86) PDBSTR (32) DNA sequence (322) RefSeq(nuc) (142) GenBank (112) EMBL (68) 3D Structure (14) PDB (14) Protein domain (14) InterPro (13) PROSITE(DOC) (1) All databases (25561)
Download RDF

	REACTION: R00432	Help
Entry	R00432 Reaction	
Name	Succinate:CoA ligase (GDP-forming)	
Definition	GTP + Succinate + CoA <=> GDP + Orthophosphate + Succinyl-CoA	
Equation	C00044 + C00042 + C00010 <=> C00035 + C00009 + C00091	
		
Reaction class	RC00002 C00035_C00044 RC00004 C00010_C00091 RC00014 C00042_C00091	
Enzyme	6.2.1.4	
Pathway	rn00020 Citrate cycle (TCA cycle) rn00640 Propanoate metabolism rn01100 Metabolic pathways rn01110 Biosynthesis of secondary metabolites rn01200 Carbon metabolism	
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate	
Orthology	K01899 succinyl-CoA synthetase alpha subunit [EC:6.2.1.4 6.2.1.5] K01900 succinyl-CoA synthetase beta subunit [EC:6.2.1.4 6.2.1.5]	
Other DBs	RHEA: 22123	

All links
Ontology (1) KEGG BRITE (1) Pathway (12) KEGG PATHWAY (10) KEGG MODULE (2) Chemical substance (6) KEGG COMPOUND (6) Chemical reaction (5) KEGG ENZYME (1) KEGG RCLASS (3) SABIO-RK (1) Gene (1628) KEGG ORTHOLOGY (2) KEGG GENES (1626) All databases (1652)
Download RDF

	REACTION: R00727	Help
Entry	R00727 Reaction	
Name	Succinate:CoA ligase (IDP-forming)	
Definition	ITP + Succinate + CoA <=> IDP + Orthophosphate + Succinyl-CoA	
Equation	C00081 + C00042 + C00010 <=> C00104 + C00009 + C00091	
		
Reaction class	RC00002 C00081_C00104 RC00004 C00010_C00091 RC00014 C00042_C00091	
Enzyme	6.2.1.4	
Pathway	rn00020 Citrate cycle (TCA cycle) rn01100 Metabolic pathways rn01110 Biosynthesis of secondary metabolites rn01120 Microbial metabolism in diverse environments rn01200 Carbon metabolism	
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate	
Orthology	K01899 succinyl-CoA synthetase alpha subunit [EC:6.2.1.4 6.2.1.5] K01900 succinyl-CoA synthetase beta subunit [EC:6.2.1.4 6.2.1.5]	

All links
Ontology (1) KEGG BRITE (1) Pathway (12) KEGG PATHWAY (10) KEGG MODULE (2) Chemical substance (6) KEGG COMPOUND (6) Chemical reaction (4) KEGG ENZYME (1) KEGG RCLASS (3) Gene (1628) KEGG ORTHOLOGY (2) KEGG GENES (1626) All databases (1651)
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KEGG

Select the human gene



ORTHOLOGY: K01899

[Help](#)

Entry	K01899	KO
Name	LSC1	
Definition	succinyl-CoA synthetase alpha subunit [EC:6.2.1.4 6.2.1.5]	
Pathway	ko00020 Citrate cycle (TCA cycle) ko00640 Propanoate metabolism ko01100 Metabolic pathways ko01110 Biosynthesis of secondary metabolites ko01120 Microbial metabolism in diverse environments ko01200 Carbon metabolism	
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate	
Disease	H00469 Mitochondrial DNA depletion syndrome	
Brite	KEGG Orthology (KO) [BR: ko00001] 09100 Metabolism 09101 Carbohydrate metabolism 00020 Citrate cycle (TCA cycle) K01899 LSC1; succinyl-CoA synthetase alpha sub 00640 Propanoate metabolism K01899 LSC1; succinyl-CoA synthetase alpha sub Enzymes [BR: ko01000] 6. Ligases 6.2 Forming carbon-sulfur bonds 6.2.1 Acid-thiol ligases 6.2.1.4 succinate---CoA ligase (GDP-forming) K01899 LSC1; succinyl-CoA synthetase alpha sul 6.2.1.5 succinate---CoA ligase (ADP-forming) K01899 LSC1; succinyl-CoA synthetase alpha sul BRITE hierarchy	
Other DBs	RN: R00432 R00727 GO: 0004776 0004775	
Genes	HSA: 8802 (SUCLG1) PTR: 470417 (SUCLG1) PPS: 100985425 (SUCLG1) GGO: 101143056 (SUCLG1) PON: 100173831 (SUCLG1) NLE: 100579244 (SUCLG1) MCC: 693991 (SUCLG1) MCF: 101867049 (SUCLG1) CSAB: 103219966 (SUCLG1) RRO: 104665951 (SUCLG1) » show all Taxonomy UniProt	

All links

Ontology (4)
KEGG BRITE (2)
GO (2)
Pathway (14)
KEGG PATHWAY (12)
KEGG MODULE (2)
Disease (1)
KEGG DISEASE (1)
Chemical reaction (6)
KEGG ENZYME (2)
KEGG REACTION (2)
KEGG RCLASS (2)
Gene (10967)
KEGG GENES (718)
KEGG MGENES (468)
RefGene (9743)
OC (38)
All databases (10992)

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KEGG

Gene information for *SUCLG1*.

Pathways/modules the gene is associated with are listed.

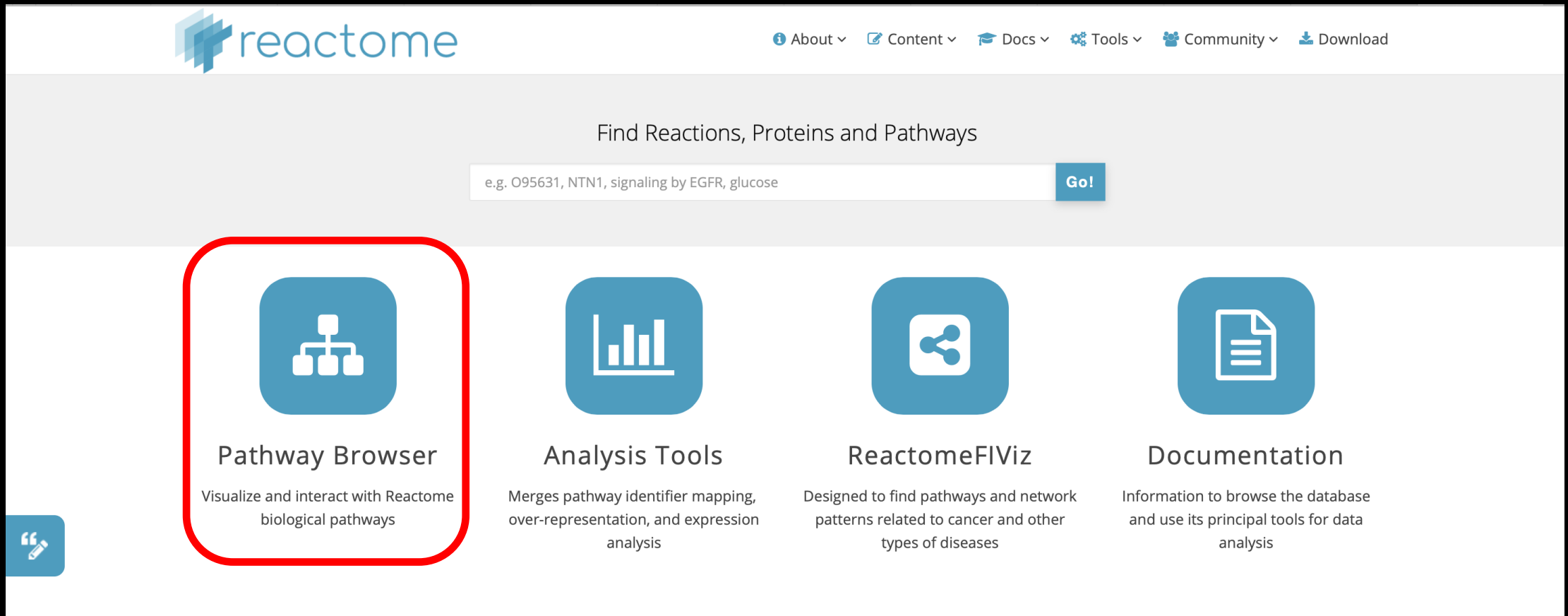
There are links to other databases.

KEGG Homo sapiens (human): 8802 Help		All links
Entry	8802 CDS T01001	Ontology (2)
Gene name	SUCLG1, GALPHA, MTDPS9, SUCLA1	KEGG BRITE (2)
Definition	(RefSeq) succinate-CoA ligase GDP/ADP-forming subunit alpha	Pathway (6)
KO	K01899 succinyl-CoA synthetase alpha subunit [EC:6.2.1.4 6.2.1.5]	KEGG PATHWAY (4)
Organism	hsa Homo sapiens (human)	KEGG MODULE (2)
Pathway	hsa00020 Citrate cycle (TCA cycle) hsa00640 Propanoate metabolism hsa01100 Metabolic pathways hsa01200 Carbon metabolism	Disease (3)
Module	hsa_M00009 Citrate cycle (TCA cycle, Krebs cycle) hsa_M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate	KEGG DISEASE (1)
Disease	H00469 Mitochondrial DNA depletion syndrome	OMIM (2)
Brite	KEGG Orthology (KO) [BR:hsa00001] 09100 Metabolism 09101 Carbohydrate metabolism 00020 Citrate cycle (TCA cycle) 8802 (SUCLG1) 00640 Propanoate metabolism 8802 (SUCLG1) Enzymes [BR:hsa01000] 6. Ligases 6.2 Forming carbon-sulfur bonds 6.2.1 Acid-thiol ligases 6.2.1.4 succinate---CoA ligase (GDP-forming) 8802 (SUCLG1) 6.2.1.5 succinate---CoA ligase (ADP-forming) 8802 (SUCLG1) BRITE hierarchy	Chemical substance (8)
SSDB	Ortholog Paralog GFIT	KEGG COMPOUND (8)
Motif	Pfam: CoA_binding Ligase_CoA_Succ_CoA_lig CoA_binding_2 Motif	Chemical reaction (4)
Other DBs	NCBI-GeneID: 8802 NCBI-ProteinID: NP_003840 OMIM: 611224 HGNC: 11449 Ensembl: ENSG00000163541 Vega: OTTHUMG00000130023 Pharos: P53597(Tbio) UniProt: P53597	KEGG ENZYME (2)
Structure	PDB: 6G4Q Thumbnails 	KEGG REACTION (2)
Position	2p11.2	Genome (1)
AA seq	346 aa AA seq DB search MTATLAAAADIATMVSGSSGLAARLLSRSFLLPQNGIRHCSYTSARQHLYVDKNKTIIC QGFTGKQGTFSHQALEYGTKLVGGTTPGKGGQTHLGLPVFNTVKEAKEQTGATASIVV PPPFAAAINEAIEAEIPLVVCITEGIPQDDMVRVKHLRLRQKTRLIGPNCPGVINPGE CKIGIMPGHIHKKGRIGIVRSGLTYEAVHQTQVGLGQSLCVGIGGDPFNGTDFIDCL EIFLNSATEGILIGEIGGNAEENAEFLQHNSGPNSKPVVSFIAGLTAPPGRMRGHA GAI IAGGKGGAKEKISALQSAGVVVSMSPAQLGTTIYKEFEKRKML	KEGG GENOME (1)
NT seq	1041 nt NT seq atgacgcgaaccccttgccgctgccgctgacatcgctaccatggctctccggcagcagcgcc ctcgccgcgcgccctgtctctgtcgcgacgcttctcctgccgcagaatggaattcgccat tgttctctacacagcttctcggcacacatctctatgttgataaaaatacaaaagattattg cagggtttcactggcacaacagggcacaccttcacagccagcaggcattggaatatggcacc aaactcgttggggaaccactccagggaaaggaggccagacacatctgggcttacctgtc tttaatactgtgaaggaggccaaagaacagacaggagcaacggctctgtcatttatgtt ctccgccttttgctgctgctgccattaatgaagctattgaggcagaaattcccttggtt gtgtgatcactgaaggaaatccccagcaggacatggtagcagtgcaagcacaactgctg cgccaggaaaagacaaggctaattgggcccaactgccctggagtcacatcaatcctggagaa tgtaaaattggcatcatgcctggccatattcacaaaaagggaaggattggcattgtgtcc agatctggcacccctgacttatgaagcagttcaccaacaacgcaagttggattggggcag tccttgtgcgttggcattggaggtgatccttttaattggaacagatttttattgactgcctc gaaatcttttgaacgattctgccacagaaggcatcatattgattggtgaaattggtggt aatgcagaagagaatgctgcagaatttttgaagcaacataattcaggtccaaattcccaag cctgtagtgtccttcattgtcgtgttaactgctcctcctgggagagaagaatgggtcatgcc ggggcaattattgctggagagaaaagggtggagctaaagagaagatctctgccttcagagt gcaggagttgtgtcagtatgtctctctgcacagctgggaaccacgatctacaaggaattt gaaaagagggaagatgctatga	Gene (21)

Reactome

reactome.org

Reactome is an online pathway database, with a focus on reactions.



The screenshot shows the Reactome website homepage. At the top is the Reactome logo and a navigation bar with links: About, Content, Docs, Tools, Community, and Download. Below the navigation bar is a search bar with the text "Find Reactions, Proteins and Pathways" and a "Go!" button. The search bar contains the example text "e.g. O95631, NTN1, signaling by EGFR, glucose". Below the search bar are four main sections, each with an icon and a description:

- Pathway Browser**: Visualize and interact with Reactome biological pathways. This section is highlighted with a red rounded rectangle.
- Analysis Tools**: Merges pathway identifier mapping, over-representation, and expression analysis.
- ReactomeFIViz**: Designed to find pathways and network patterns related to cancer and other types of diseases.
- Documentation**: Information to browse the database and use its principal tools for data analysis.

There is also a small blue icon in the bottom left corner of the page.

Reactome

<https://reactome.org/PathwayBrowser/>

The screenshot displays the Reactome Pathway Browser interface. At the top, the Reactome logo and version (3.7) are shown. The main navigation bar includes a search bar, a dropdown for "Pathways for: Homo sapiens", and buttons for Citation, Analysis, Tour, and Layout. On the left, the "Event Hierarchy" panel lists various biological processes, with "Metabolism" highlighted. The central area shows a complex network of pathways, with "Retinoid metabolism and transport" selected and highlighted in a blue box. Below the network, a tabbed interface shows "Description", "Molecules", "Structures", "Expression", "Analysis", and "Downloads". A description box at the bottom explains that the interface displays details for selected items, including input/output molecules, summary, references, and supporting evidence.

reactome 3.7

Pathways for: Homo sapiens

Citation: Analysis: Tour: Layout:

Search for a term, e.g. pten ...

Event Hierarchy:

- Autophagy
- Cell Cycle
- Cell-Cell communication
- Cellular responses to external stimuli
- Chromatin organization
- Circadian Clock
- Digestion and absorption
- Developmental Biology
- Disease
- DNA Repair
- DNA Replication
- Extracellular matrix organization
- Gene expression (Transcription)
- Hemostasis
- Immune System
- Metabolism
- Metabolism of proteins
- Metabolism of RNA
- Muscle contraction
- Neuronal System
- Organelle biogenesis and maintenance
- Programmed Cell Death
- Protein localization
- Reproduction
- Sensory Perception
- Signal Transduction
- Transport of small molecules
- Vesicle-mediated transport

Muscle contraction

Digestion and absorption

Circadian Clock

Immune System

Metabolism of RNA

Chromatin organization

DNA Repair

DNA Replication

Cell Cycle

Programmed Cell Death

Transport of small molecules

Reproduction

Cellular responses to external stimuli

Organelle biogenesis and maintenance

Autophagy

Protein localization

Extracellular matrix organization

Cell-Cell communication

Vesicle-mediated transport

Metabolism of proteins

Disease

Gene expression (Transcription)

Hemostasis

Neuronal System

Developmental Biology

Signal Transduction

Retinoid metabolism and transport

Sensory Perception

Description

Molecules

Structures

Expression

Analysis

Downloads

Displays details when you select an item in the Pathway Browser. For example, when a reaction is selected, shows details including the input and output molecules, summary and references containing supporting evidence. When relevant, shows details of the catalyst, regulators, preceding and following events.

Reactome



[About](#) [Content](#) [Docs](#) [Tools](#) [Community](#) [Download](#)

Find Reactions, Proteins and Pathways

e.g. O95631, NTN1, signaling by EGFR, glucose

Go!



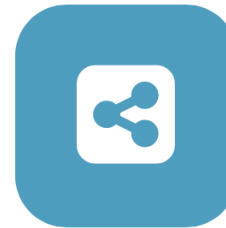
Pathway Browser

Visualize and interact with Reactome biological pathways



Analysis Tools

Merges pathway identifier mapping, over-representation, and expression analysis



ReactomeFIViz

Designed to find pathways and network patterns related to cancer and other types of diseases



Documentation

Information to browse the database and use its principal tools for data analysis



Reactome

Input:

SENP7
MOB1B
CARMIL1
PRRC2A
TERF2
RFWD3
PARP1
POT1
ATM
MPHOSPH6

 Analysis tools


Analyse gene list


Analyse gene expression


Species Comparison


Tissue Distribution

Reactome v76

Click to learn more about our analysis tools

Your dataOptionsAnalysis

Step 1: Select a file from your computer or paste your own data and click on the corresponding "Continue" button.

Select data file for analysis: no file selected

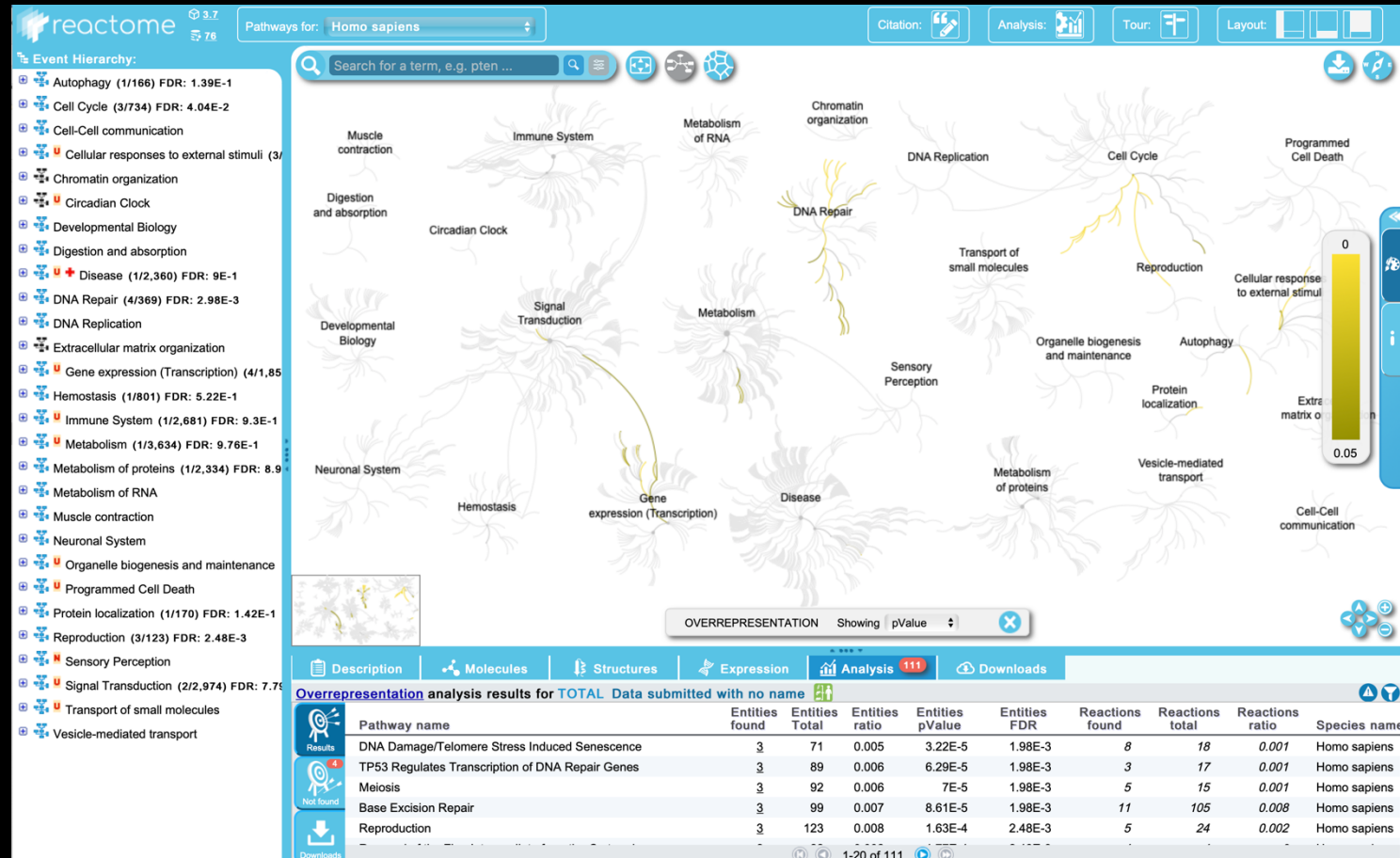
Select a file to analyse

Paste your data to analyse or try example data sets:
Paste your data here or select an example from the right >>

Some examples:

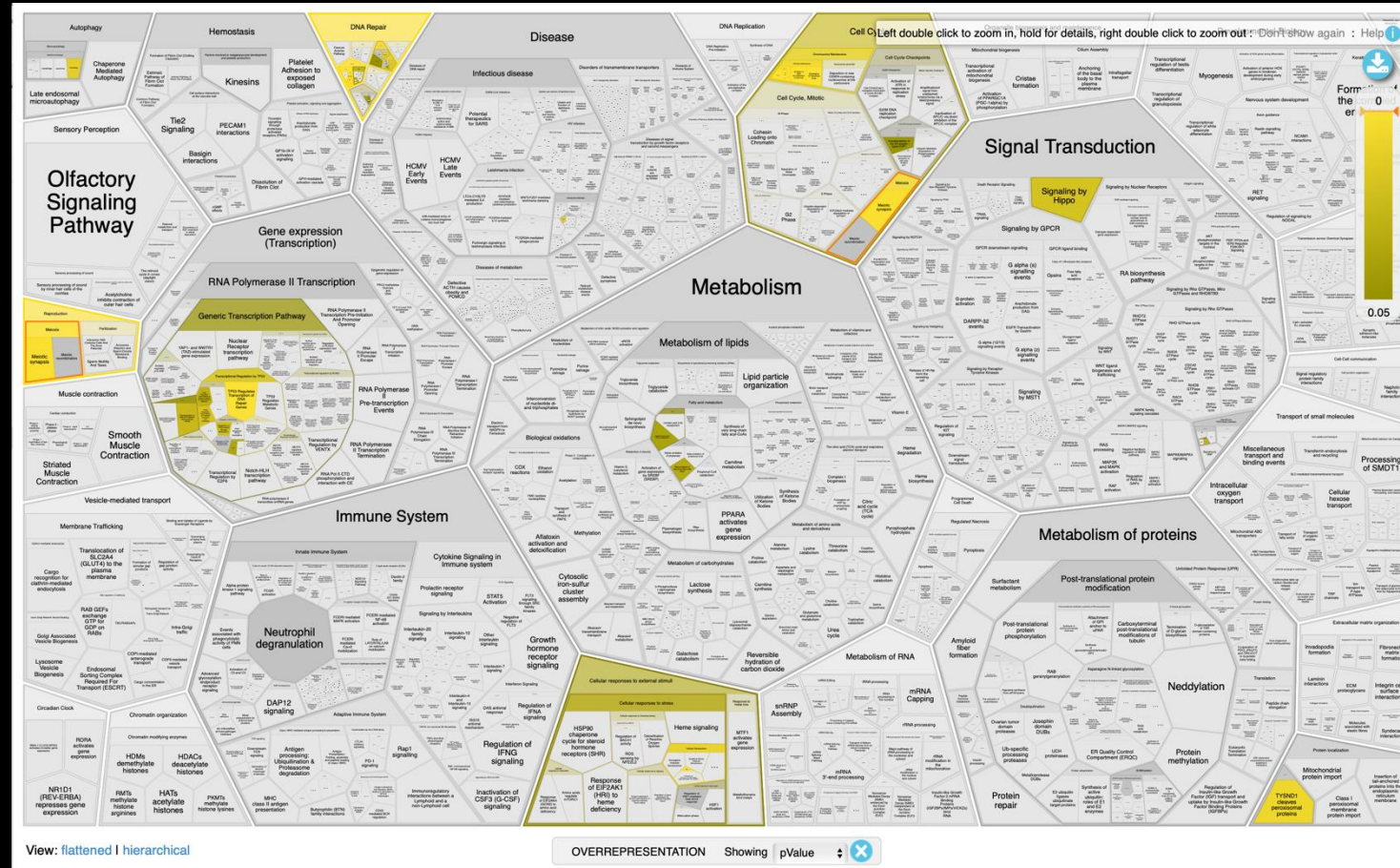
Reactome

Results



Reactome

Results

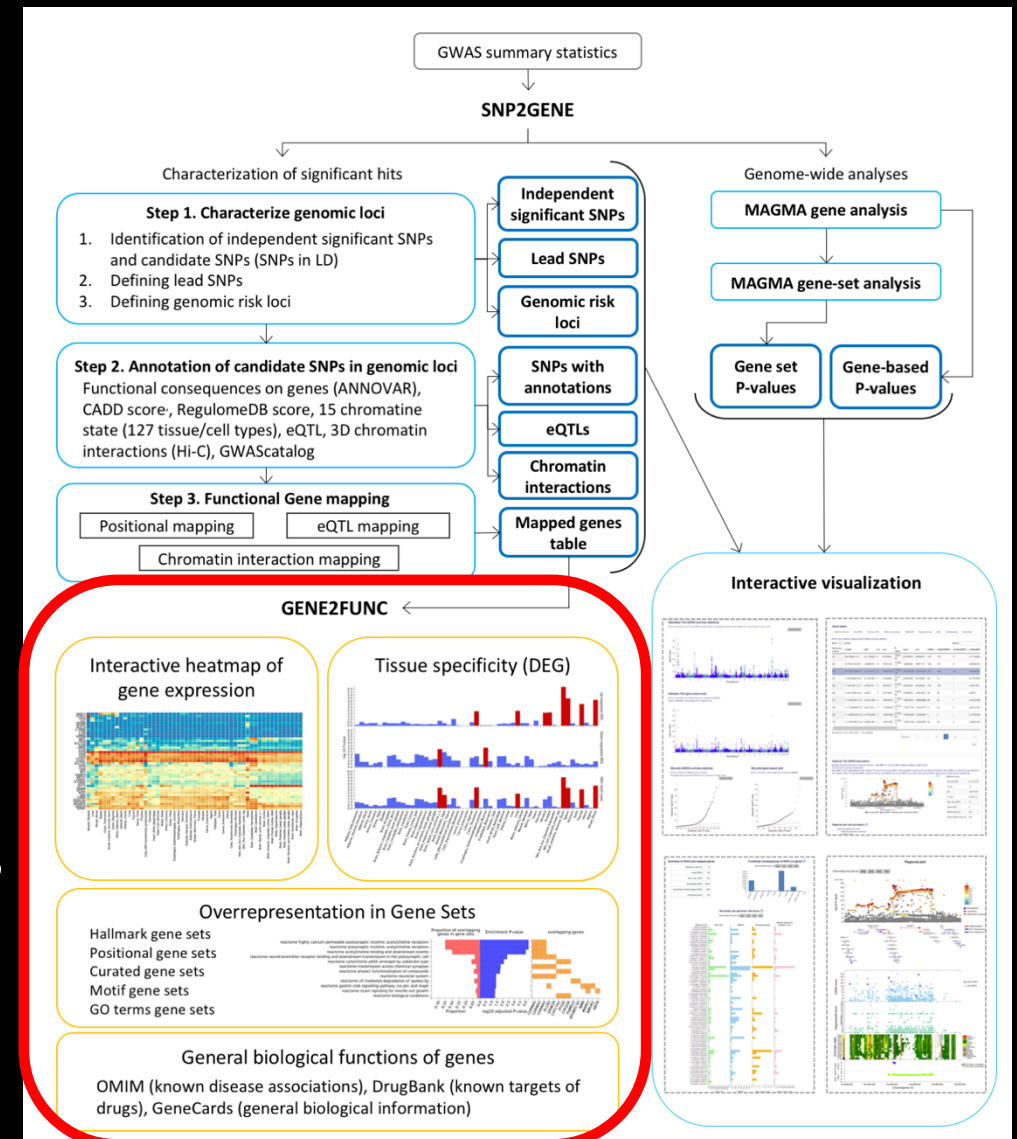


Gene set/pathway analysis in FUMA

Recall that FUMA's SNP2GENE feature takes GWAS summary statistics and outputs a list of mapped genes.

The GENE2FUNC feature takes a set of genes as input and can perform overrepresentation analyses.

Gene sets include: Gene Ontologies, KEGG pathways, Reactome pathways, and DE genes across various tissues/cell types



Gene set analysis in FUMA – GENE2FUNC

Provide a list of genes (copy and paste or upload a file).

You need to specify a “background” set of genes to compare your list with.

FUMA lets you select several different gene sets defined by gene expression profiles of various tissues.

You can also choose one of several multiple-testing correction methods and a “significance” threshold.

The screenshot shows the top part of the FUMA GENE2FUNC web interface. It is divided into two main columns: 'Genes of interest' and 'Background genes'.

Genes of interest:

- Instruction: "Paste or upload a file that contains gene-symbols. Priority is given to the text box if both fields are used."
- Section 1: "Paste genes" with a text area for "Please enter each gene per line here."
- Section 2: "Upload file" with a "Choose File" button and the text "no file selected".
- Red instruction box: "Please either copy-paste or upload a list of genes to test."

Background genes:

- Instruction: "Specify background gene-set. This will be used in the hypergeometric test."
- Section 1: "Select background genes by gene-type" with a "Clear" button and a list: "All", "Protein coding", "lncRNA", "ncRNA", "Processed transcripts".
- Section 2: "Paste custom list of background genes" with a text area for "Please enter each gene per line here."
- Section 3: "Upload a file with a custom list of background genes" with a "Choose File" button and the text "no file selected".
- Red instruction box: "Please provide background genes."

Other optional parameters:

- Ensembl version: dropdown menu set to "v92".
- Custom gene set files: "add file" button with instruction "File is required to have GMT format with an extension *.gmt".
- Gene expression data sets: dropdown menu with options: "GTEx v8: 54 tissue types", "GTEx v8: 30 general tissue types", "GTEx v7: 53 tissue types", "GTEx v7: 30 general tissue types".
- ☐ Exclude the MHC region.
- Desired multiple test correction method for gene-set enrichment testing: dropdown menu set to "Benjamini-Hochberg (FDR)".
- Maximum adjusted P-value for gene set association (<): input field set to "0.05".
- Minimum overlapping genes with gene-sets (>): input field set to "2".
- Title: input field with an "Optional" icon.
- "Submit" button.

This screenshot shows the 'Other optional parameters' section of the FUMA GENE2FUNC web interface, with a dropdown menu open for the 'Gene expression data sets' field.

Other optional parameters:

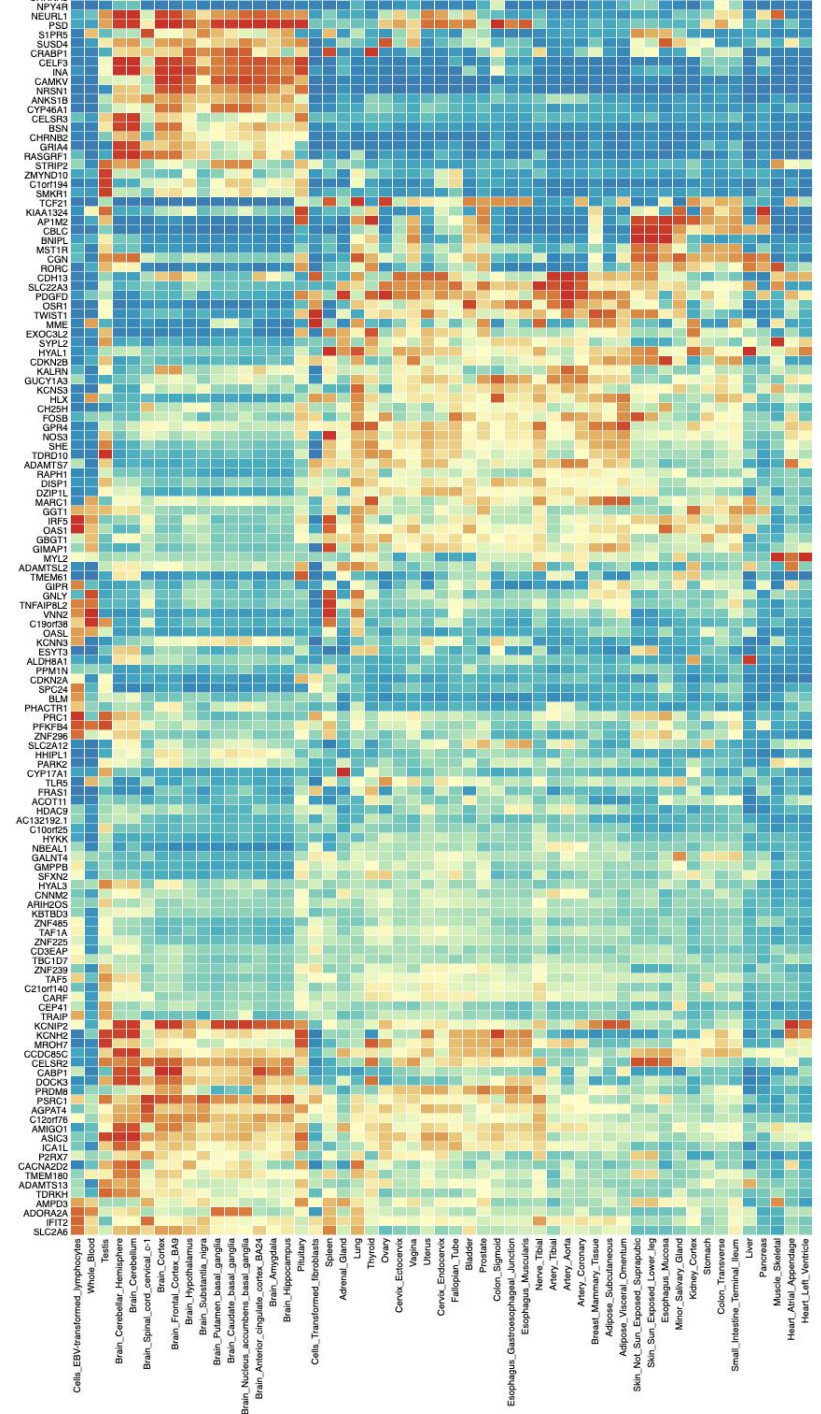
- Ensembl version: dropdown menu set to "v92".
- Custom gene set files: "add file" button with instruction "File is required to have GMT format with an extension *.gmt".
- Gene expression data sets: dropdown menu is open, showing options: "GTEx v8: 54 tissue types", "GTEx v8: 30 general tissue types", "GTEx v7: 53 tissue types", "GTEx v7: 30 general tissue types".
- ☐ Exclude the MHC region.
- Desired multiple test correction method for gene-set enrichment testing: dropdown menu is open, showing options: "Bonferroni", "Sidak", "Holm-Sidak", "Holm", "Simes-Hochberg", "Hommel", "Benjamini-Hochberg (FDR)" (which is selected), "Benjamini-Yekutieli (FDR)", "two-step Benjamini-Hochberg (FDR)", "two-step Benjamini-Krieger-Yekutieli (FDR)".
- Maximum adjusted P-value for gene set association (<): input field set to "0.05".
- Minimum overlapping genes with gene-sets (>): input field set to "2".
- Title: input field with an "Optional" icon.
- "Submit" button.

GENE2FUNC

Input: 644 genes from the hw4 data.

Output (first part):
gene expression heatmap

https://fuma.ctglab.nl/browse/179#g2f_summaryPanel



Output (second part)
Differential expression of genes across tissues
(Are the genes you supplied enriched for
DE genes in particular tissues?)

Figure 2 displays three bar charts showing the $-\log_{10}$ P-values for 100 differentially expressed genes across three comparisons: Up-regulated DEO, Down-regulated DEO, and DEO (both sides). The y-axis for all charts is $-\log_{10}$ P-value, ranging from 0 to 6. The x-axis lists 100 genes. The top chart (Up-regulated DEO) shows high significance for genes like Brain_Cerebellar_Hemisphere and Brain_Pituitary. The middle chart (Down-regulated DEO) shows high significance for genes like Brain_Cerebellar_Hemisphere and Brain_Pituitary. The bottom chart (DEO (both sides)) shows high significance for genes like Brain_Cerebellar_Hemisphere and Brain_Pituitary.

GENE2FUNC

Overrepresentation of your genes
in pre-defined gene sets

Enrichment of input genes in Gene Sets

i Plots and tables only display gene sets with adjusted P-value < 0.05. When adjusted P-value threshold is set to > 0.05, all results passed threshold are included in the GS.txt field downloadable from "Summary" tab.

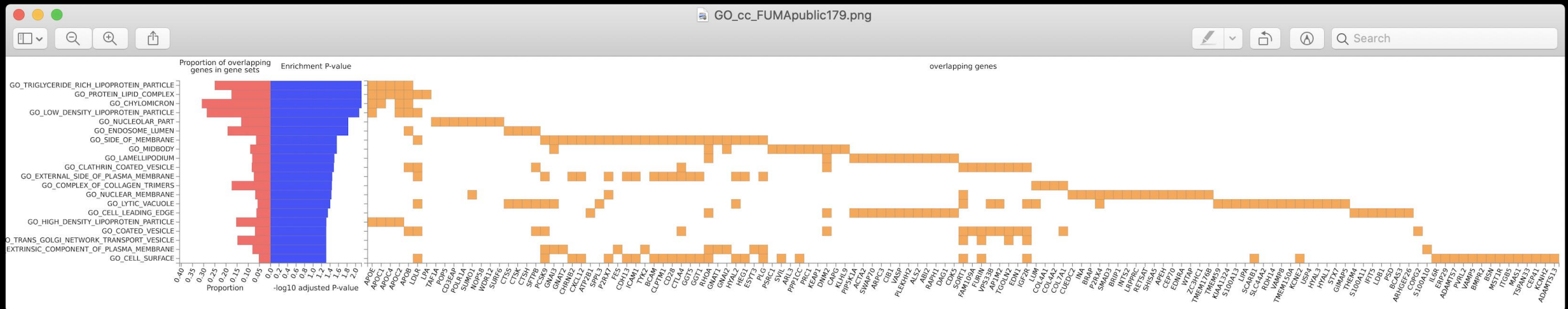
i If there is no significant gene sets (adjusted P-value < 0.05) in user provided custom gene sets, they are not displayed in this page, but all results passed threshold are included in the GS.txt field downloadable from "Summary" tab.

Hallmark gene sets (MsigDB h)	(3)
Positional gene sets (MsigDB c1)	(28)
Curated_gene_sets	(39)
Chemical and Genetic pertubation gene sets (MsigDB c2)	(17)
All Canonical Pathways (MsigDB c2)	(21)
BioCarta (MsigDB c2)	(1)
KEGG (MsigDB c2)	(2)
Reactome (MsigDB c2)	(16)
microRNA targets (MsigDB c3)	(0)
TF targets (MsigDB c3)	(3)
All computational gene sets (MsigDB c4)	(11)
Cancer gene neighborhoods (MsigDB c4)	(3)
Cancer gene modules (MsigDB c4)	(9)
GO biological processes (MsigDB c5)	(124)
GO cellular components (MsigDB c5)	(20)
GO molecular functions (MsigDB c5)	(17)
Oncogenic signatures (MsigDB c6)	(1)
Immunologic signatures (MsigDB c7)	(4)
WikiPathways	(12)
GWAS catalog reported genes	(152)

GENE2FUNC

For example, choose “GO cellular components”

You can see exactly which genes in our list overlapped with these 20 GO terms, and how significant the overlap/enrichment is.



GENE2FUNC

Results are also viewable as a table.

GO cellular components (MsigDB c5) (20)

Plot / Table

GeneSet	N	n	P-value	adjusted P	genes
GO_TRIGLYCERIDE_RICH_LIPOPROTEIN_PARTICLE	20	5	3.06e-5	7.23e-3	APOE, APOC1, APOC4, APOC2, APOB
GO_PROTEIN_LIPID_COMPLEX	40	7	3.71e-5	7.23e-3	LDLR, APOE, APOC1, APOC4, APOC2, APOB, LPA
GO_CHYLOMICRON	13	4	3.74e-5	7.23e-3	APOE, APOC1, APOC2, APOB
GO_LOW_DENSITY_LIPOPROTEIN_PARTICLE	14	4	5.66e-5	8.21e-3	LDLR, APOE, APOC2, APOB
GO_NUCLEOLAR_PART	61	8	1.49e-4	1.49e-2	TAF1A, POP5, CD3EAP, POLR1A, SUMO1, NOP58, WDR12, SURF6
GO_ENDOSOME_LUMEN	26	5	1.54e-4	1.49e-2	CTSS, CTSK, CTSH, APOB, SFTPB
GO_SIDE_OF_MEMBRANE	402	26	3.35e-4	2.76e-2	PCSK9, GNAI3, GNAT2, CHRN2, CXCL12, ATP2B1, SPPL3, P2RX7, FES, CDH13, ICAM1, TYK2, LDLR, BCAM, CLPTM1, CD28, CTLA4, GGT5, GGT1, RHOA, GNAT1, GNAI2, HYAL2, HEG1, ESYT3, PLG
GO_MIDBODY	132	12	3.81e-4	2.76e-2	PSRC1, GNAI3, SVIL, ARL3, PPP1CC, PRC1, KEAP1, DN2M, CAPG, RHOA, GNAI2, KLHL9
GO_LAMELLIPODIUM	171	14	4.90e-4	3.16e-2	PIP5K1A, ACTA2, SWAP70, ARPC3, CIB1, DN2M, VASP, PLEKHH2, ALS2, ABI2, RAPH1, RHOA, DAG1, CDK5
GO_CLATHRIN_COATED_VESICLE	155	13	5.58e-4	3.24e-2	SORT1, FAM109A, FURIN, VPS33B, AP1M2, DN2M, LDLR, APOB, TGOLN2, SFTPB, CTLA4, EDN1, IGF2R
GO_EXTERNAL_SIDE_OF_PLASMA_MEMBRANE	214	16	6.70e-4	3.53e-2	PCSK9, CHRN2, CXCL12, P2RX7, CDH13, ICAM1, LDLR, BCAM, CLPTM1, CD28, CTLA4, GGT5, GGT1, HYAL2, HEG1, PLG
GO_COMPLEX_OF_COLLAGEN_TRIMERS	23	4	7.46e-4	3.61e-2	LUM, COL4A1, COL4A2, COL7A1
GO_NUCLEAR_MEMBRANE	278	19	8.29e-4	3.70e-2	SORT1, CUEDC2, INA, BRAP, P2RX7, P2RX4, SMAD3, BRIP1, INTS2, LRPPRC, RETSAT, SUMO1, SHISA5, APEH, CEP70, EDNRA, WTAP, ZC3HC1, TMEM176B
GO_LYTIC_VACUOLE	520	30	9.54e-4	3.95e-2	TMEM59, PCSK9, KIAA1324, SORT1, GNAI3, CTSS, CTSK, S100A13, LIPA, LUM, SPPL3, P2RX4, SCARB1, CTSH, VPS33B, AP1M2, SLC44A2, LDLR

One other FUMA function

Recall that FUMA uses GWAS summary statistics to perform gene-based testing in MAGMA.

MAGMA can integrate the gene-based p-values into a gene set testing framework.

Essentially this tests whether genes in a gene set are more strongly associated with the phenotype (compared to other genes).

Recall that this uses ALL genes, not just a set you chose.

MAGMA results

Gene sets defined by differential expression
in various tissues.

MAGMA Tissue Expression Analysis

i MAGMA gene-property analysis is performed for gene expression of user selected data sets.

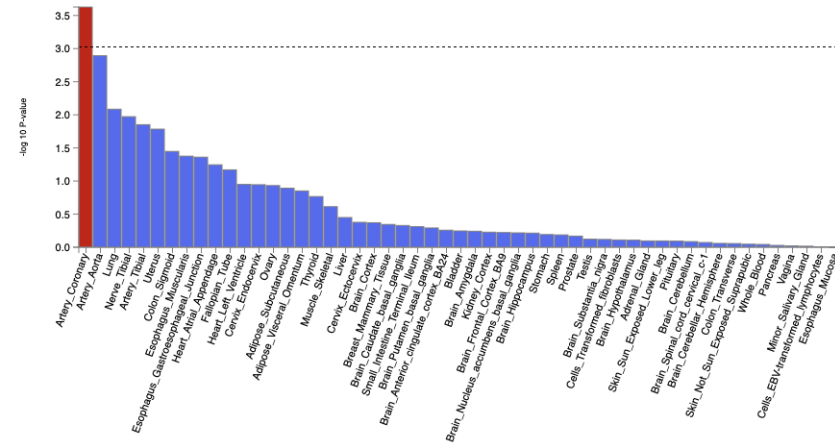
Details of gene expression data sets are available at [Tutorial](#). Full results are downloadable from "Download" tab.

Note that MAGMA gene-property analyses uses the full distribution of SNP p-values and is different from an enrichment test of DEG (differentially expressed genes) as implemented in GENE2FUNC that only tests for enrichment of prioritised genes.

Order tissue by :

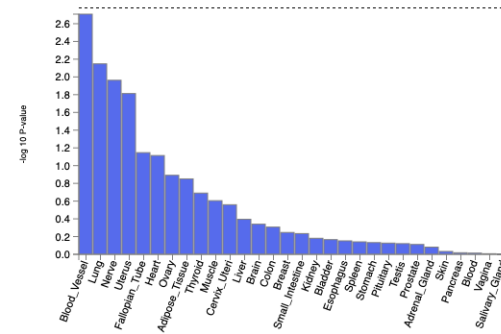
GTEx v7 53 tissue types

Download the plot as



GTEx v7 30 general tissue types

Download the plot as



GREAT

Genomic Regions Enrichment of Annotations

Publication

<http://bejerano.stanford.edu/papers/GREAT.pdf>

Website

<http://great.stanford.edu/public/html/index.php>

More documentation

<https://great-help.atlassian.net/wiki/spaces/GREAT/overview?homepageId=655450>

GREAT

Focus on “b”

Scenario: you want to know if some regions identified by your study are enriched for an annotation.

Arrows = gene transcription start sites

Black arrows = for genes without a specific annotation of interest

Green arrows = for genes without the annotation

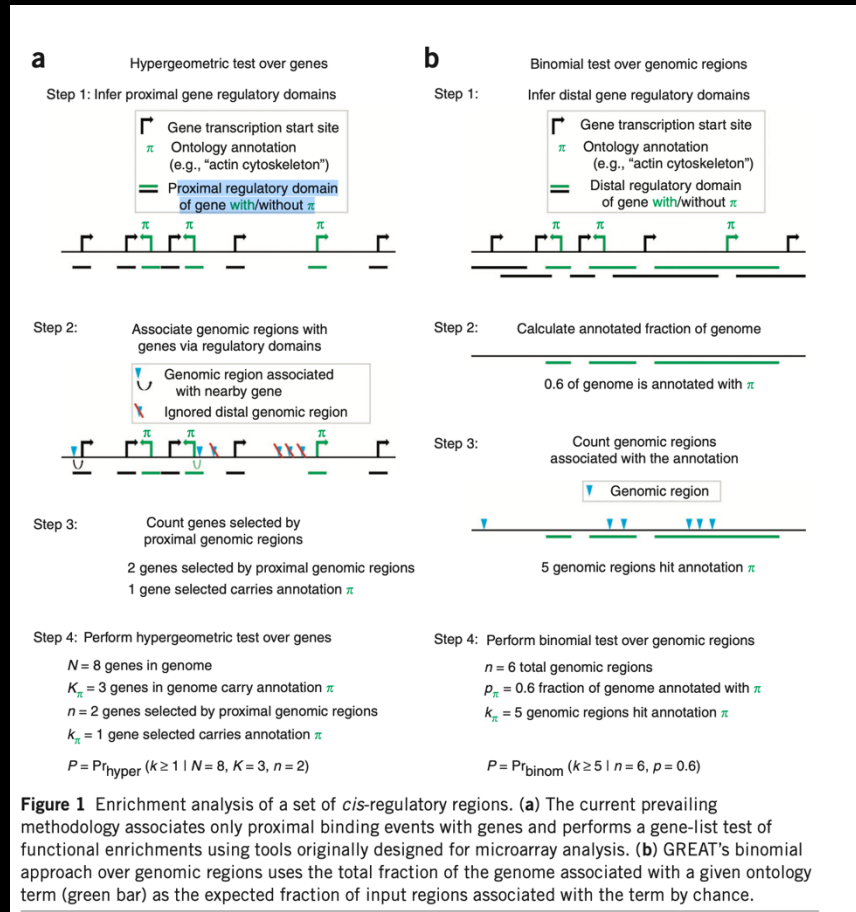
Black lines = regulatory regions for gene without the annotation

Green arrows = regulatory regions for genes with the annotation

Calculate the fraction of the genome overlapping the green lines.

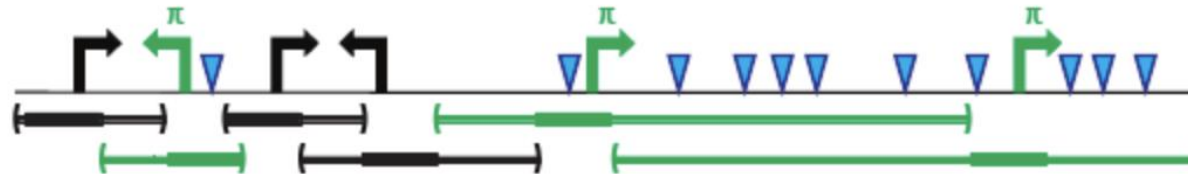
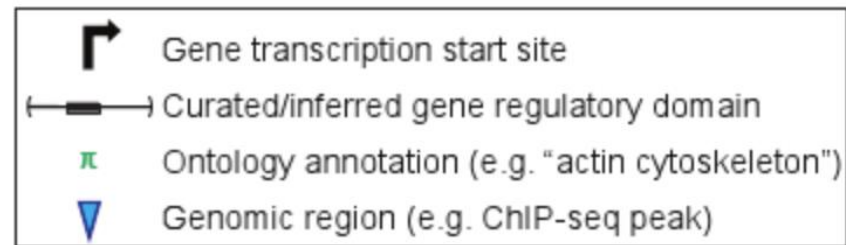
How many of your genomic regions overlap the green lines?

Perform a binomial test.



GREAT

Example



Binomial test over genomic regions

$n = 11$ total genomic regions
 $p_{\pi} = 0.75$ fraction of genome annotated with π
 $k_{\pi} = 11$ genomic regions annotated with π
P-value = 0.04

Hypergeometric test over genes

$N = 6$ total genes
 $K_{\pi} = 3$ genes annotated with π
 $n = 3$ genes with an associated genomic region
 $k_{\pi} = 3$ genes annotated and with a genomic region
P-value = 0.05

GREAT

Select a species and provide input in BED format

GREAT

OverviewNewsUse GREATDemoVideoHow to CiteHelpForum

GREAT version 4.0.4current (08/19/2019 to now)

GREAT predicts functions of cis-regulatory regions.

Many coding genes are well annotated with their biological functions. Non-coding regions typically lack such annotation. GREAT assigns biological meaning to a set of non-coding genomic regions by analyzing the annotations of the nearby genes. Thus, it is particularly useful in studying cis functions of sets of non-coding genomic regions. Cis-regulatory regions can be identified via both experimental methods (e.g. [ChIP-seq](#)) and by computational methods (e.g. [comparative genomics](#)). For more see our [Nature Biotech Paper](#).

News

- Aug. 19, 2019: GREAT version 4 [adds support for human hg38 assembly and updates ontology datasets for all supported assemblies](#).
- Sep. 8, 2018: GREAT has served over 1 million job submissions.
- Oct. 23, 2017: GREAT is moved to a VM to eliminate proxy errors.
- June 22, 2017: GREAT hardware upgrade to meet increasing submission volume.
- Nov. 16, 2015: The [GREAT user help forums](#) are frozen.
- Feb. 15, 2015: GREAT version 3 [switches to Ensembl genes, adds support for zebrafish danRer7 and mouse mm10 assemblies, and adds new ontologies](#).
- Apr. 3, 2012: GREAT version 2 [adds new annotations to human and mouse ontologies and visualization tools for data exploration](#).
- Feb. 18, 2012: The [GREAT user help forums](#) are opened.
- May 2, 2010: GREAT version 1 is launched, concurrent to [Nature Biotechnology publication](#) (reprint, Faculty of 1000 "Must Read"). [How to Cite GREAT?](#)

[More news items...](#)

Species Assembly

☐ Human: GRCh38 ([UCSC hg38, Dec. 2013](#))

☐ Human: GRCh37 ([UCSC hg19, Feb. 2009](#))

☐ Mouse: GRCh38 ([UCSC mm10, Dec. 2011](#))

☐ Mouse: NCBI build 37 ([UCSC mm9, Jul. 2007](#))

[Can I use a different species or assembly?](#)

Test regions

☒ BED file:

Choose File...no file selected

☐ BED data:

[What should my test regions file contain?](#)

[How can I create a test set from a UCSC Genome Browser annotation track?](#)

Background regions

☒ Whole genome

☐ BED file:

Choose File...no file selected

☐ BED data:

[When should I use a background set?](#)

[What should my background regions file contain?](#)

Association rule settings

Show settings »

SubmitReset

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GREAT

Example: plug in a list of 235 SNPs from a GWAS of height
(<http://great.stanford.edu/public/html/demo.php>)

 [Overview](#) [News](#) [Use GREAT](#) [Demo](#) [Video](#) [How to Cite](#) [Help](#) [Forum](#)

GREAT version 4.0.4 current (08/19/2019 to now) 

Choose one of the data sets below to run. In each case, the data set is run against a whole genome background, using the *basal plus extension* genomic association rule with its default values.

Set	Experiment	Set size	Species	Cell/tissue type	Reference	GREAT Enrichment Highlights
☐ SRF (activator)	ChIP-seq	556	Human (hg19)	Jurkat	Valouev et al. 2008	Actin-related terms
☐ NRSF/REST (repressor)	ChIP-seq	1,712	Human (hg19)	Jurkat	Valouev et al. 2008	Neurotransmitters and ion channels
☐ p300 transcriptional co-activator	ChIP-seq	2,105	Mouse (mm10)	Embryonic limb E11.5	Visel et al. 2009	Embryonic limb development
☐ Ultraconserved elements	Comparative genomics	481	Human (hg19)	N/A	Bejerano et al. 2004	Transcription regulation and developmental defects
☒ Height-associated unlinked SNPs	GWAS	235	Human (hg19)	N/A	NHGRI compendium	Bone growth terms

Submit

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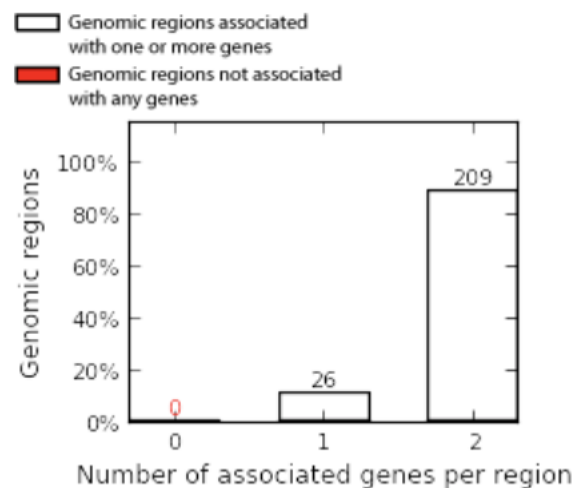
GREAT

+ Region-Gene Association Graphs

What do these graphs illustrate?

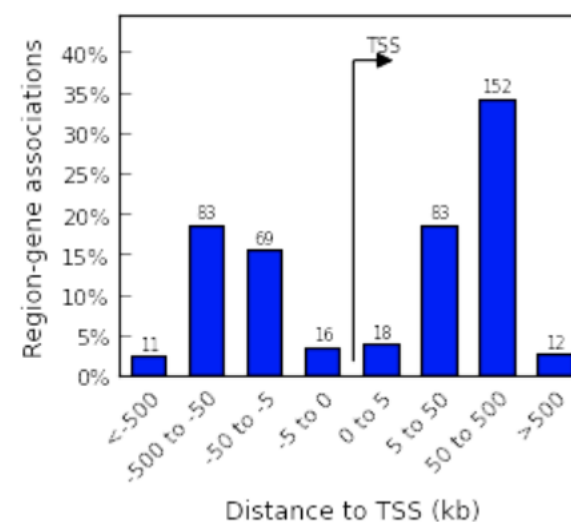
Number of associated genes per region

[Download as PDF.](#)



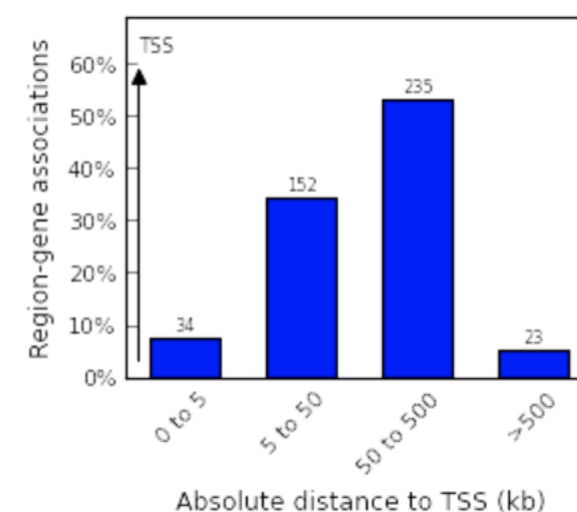
Binned by orientation and distance to TSS

[Download as PDF.](#)



Binned by absolute distance to TSS

[Download as PDF.](#)



GREAT

Output controls

Global Controls

Global Export

Which data is exported by each option?

Ontologies

Show Non-Empty

Expand All

Collapse All

GO

☒ GO Molecular Function

☒ GO Biological Process

☒ GO Cellular Component

Click a link above to jump to the corresponding table.

What data does each ontology provide?

Can I use other ontologies?

Phenotype

☒ Mouse Phenotype

☒ Mouse Phenotype Single KO

☒ Human Phenotype

Genes

☒ Ensembl Genes

Columns

Show All

Hide All

General

☐ Ontology Term ID

Binomial over Regions

☒ Rank

☒ Raw P-Value

☐ Bonferroni P-Value

☒ FDR Q-Value

☒ Fold Enrichment (Obs/Exp)

☐ Expected (n * p)

☒ Observed Region Hits (k)

☐ Genome Fraction (p)

☒ Region Set Coverage (k/n)

Hypergeometric over Genes

☒ Rank

☐ Raw P-Value

☐ Bonferroni P-Value

☒ FDR Q-Value

☒ Fold Enrichment (Obs/Exp)

☐ Expected (n * K/N)

☒ Observed Gene Hits (k)

☒ Total Genes (K)

☒ Gene Set Coverage (k/n)

☐ Term Gene Coverage (k/K)

Click on a link above to sort all tables by the named column.

What do each of the columns mean?

Top Rows/Table

20

Set

Default

Minimum Region-based Fold Enrichment

2

Set

Default

Observed Gene Hits

1

Set

Default

Term Name Filter

Filter

Clear

Where is my term?

Term Annotation Count

Min: 1

Max: 1000

Set

Default

What is the term annotation count?

Statistical Significance

☐ Raw P-Value

☐ Bonferroni

☒ FDR

Threshold: 0.05

Set

Default

Significant P-Values are **bold** in the tables below.

Terms are **bold** when significant by both binomial and hypergeometric tests.

Which statistics does GREAT provide?

View

☒ Significant By Both

☐ Significant By Region-based Binomial

☐ Ignore statistical significance

Which view should I use?

Reset view

Example output for GO biological process

GO Biological Process (20+ terms)												Global controls
Table controls: Export ▾ Shown top rows in this table: 20 Set Term annotation count: Min: 1 Max: Inf Set Visualize this table: [select one] ▾												
Term Name	Binom Rank	Binom Raw P-Value ▲	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
oncogene-induced cell senescence	3	1.7272e-18	7.5682e-15	128.5039	10	4.26%	75	4.1815e-2	64.6307	2	2	0.70%
skeletal system development	28	1.7143e-11	8.0479e-9	3.0246	46	19.57%	1	5.5453e-7	4.1697	30	465	10.45%
negative regulation of nitrogen compound metabolic process	36	5.2151e-11	1.9042e-8	2.0717	79	33.62%	12	3.3690e-3	1.9925	48	1,557	16.72%
chondrocyte differentiation	40	8.0475e-11	2.6446e-8	6.4496	20	8.51%	3	1.8272e-4	10.2588	10	63	3.48%
negative regulation of nucleobase-containing compound metabolic process	55	2.8821e-10	6.8883e-8	2.0720	74	31.49%	33	6.1454e-3	1.9640	43	1,415	14.98%
negative regulation of gene expression	61	4.8352e-10	1.0419e-7	2.0340	75	31.91%	17	3.4228e-3	2.0034	46	1,484	16.03%
skeletal system morphogenesis	62	4.8529e-10	1.0289e-7	4.0645	28	11.91%	2	1.9009e-4	5.2570	17	209	5.92%
cartilage development	66	6.2737e-10	1.2495e-7	4.4829	25	10.64%	5	2.7834e-4	5.9528	14	152	4.88%
negative regulation of RNA metabolic process	68	7.7255e-10	1.4934e-7	2.1220	68	28.94%	41	8.0168e-3	2.0068	39	1,256	13.59%
chondrocyte differentiation involved in endochondral bone morphogenesis	83	3.9875e-9	6.3152e-7	17.4059	9	3.83%	7	6.8854e-4	29.3776	5	11	1.74%
cartilage development involved in endochondral bone morphogenesis	97	1.3825e-8	1.8735e-6	12.3285	10	4.26%	10	1.3848e-3	16.8602	6	23	2.09%
endochondral bone morphogenesis	98	1.3953e-8	1.8715e-6	7.2616	14	5.96%	4	2.5376e-4	11.4054	9	51	3.14%
negative regulation of transcription, DNA-templated	103	1.7062e-8	2.1775e-6	2.0619	62	26.38%	36	6.7704e-3	2.0830	37	1,148	12.89%
connective tissue development	105	1.8601e-8	2.3287e-6	3.6482	26	11.06%	8	7.8921e-4	4.8963	15	198	5.23%
negative regulation of nucleic acid-templated transcription	117	3.4671e-8	3.8953e-6	2.0229	62	26.38%	45	1.1910e-2	2.0078	37	1,191	12.89%
negative regulation of RNA biosynthetic process	119	4.4820e-8	4.9509e-6	2.0088	62	26.38%	52	1.3639e-2	1.9812	37	1,207	12.89%
response to growth hormone	120	4.9684e-8	5.4425e-6	10.7355	10	4.26%	65	3.3361e-2	9.5045	5	34	1.74%
positive regulation of fibroblast proliferation	131	1.2924e-7	1.2968e-5	7.3800	12	5.11%	64	3.3341e-2	7.3167	6	53	2.09%
embryonic morphogenesis	151	4.1544e-7	3.6165e-5	2.2891	42	17.87%	63	3.0900e-2	2.4588	21	552	7.32%
morphogenesis of a branching epithelium	183	3.4599e-6	2.4853e-4	3.2088	21	8.94%	22	4.2208e-3	4.8778	12	159	4.18%

The test set of 235 genomic regions picked 287 (2%) of all 18,549 genes.
 GO Biological Process has 13,145 terms covering 16,621 (90%) of all 18,549 genes, and 1,251,831 term - gene associations.
 13,145 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Other enrichment categories

GO Cellular Component (1 term)Global controls

Table controls: Export Shown top rows in this table: 20 Set Term annotation count: Min: 1 Max: Inf Set Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
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proteinaceous extracellular matrix	9	3.0842e-8	5.9148e-6	2.9578	33	14.04%	1	2.3838e-2	3.2315	18	360	6.27%
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The test set of 235 genomic regions picked 287 (2%) of all 18,549 genes.
GO Cellular Component has 1,726 terms covering 17,726 (96%) of all 18,549 genes, and 380,581 term - gene associations.
1,726 ontology terms (100%) were tested using an annotation count range of [1, Inf].

GO Molecular Function (no terms)Global controls

Human Phenotype (20+ terms)Global controls

Table controls: Export Shown top rows in this table: 20 Set Term annotation count: Min: 1 Max: Inf Set Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
-----------	------------	-------------------	-----------------	-----------------------	----------------------------	---------------------------	------------	-----------------	-----------------------	--------------------------	-------------------	-------------------------

Growth delay	2	1.8159e-10	6.0578e-7	2.5886	52	22.13%	4	1.8731e-4	2.6802	35	844	12.20%
Aplasia/hypoplasia affecting bones of the axial skeleton	6	9.8631e-9	1.0968e-5	3.1774	32	13.62%	25	5.5121e-3	3.1357	18	371	6.27%
Growth abnormality	7	1.1452e-8	1.0915e-5	2.1202	60	25.53%	14	7.1577e-4	2.2064	41	1,201	14.29%
Short stature	8	1.6792e-8	1.4004e-5	2.7019	39	16.60%	1	6.2158e-5	3.3201	30	584	10.45%
Abnormality of the rib cage	10	4.1760e-8	2.7863e-5	4.4357	20	8.51%	10	5.2829e-4	5.0549	14	179	4.88%
Aplasia/hypoplasia involving the skeleton	15	9.0336e-8	4.0181e-5	2.5752	38	16.17%	11	5.6075e-4	3.0315	25	533	8.71%
Abnormality of body height	16	1.0135e-7	4.2264e-5	2.5228	39	16.60%	2	4.0652e-5	3.2006	31	626	10.80%
Hypoplastic iliac wing	29	6.3629e-7	1.4639e-4	11.5678	8	3.40%	75	4.1556e-2	10.7718	4	24	1.39%
Abnormality of lower limb epiphysis morphology	31	8.4191e-7	1.8120e-4	11.1381	8	3.40%	62	2.3452e-2	8.9765	5	36	1.74%
Short ribs	33	8.9185e-7	1.8032e-4	9.1222	9	3.83%	79	4.3026e-2	7.5152	5	43	1.74%
Abnormality of limb epiphysis morphology	34	9.0304e-7	1.7721e-4	6.8411	11	4.68%	41	1.5263e-2	6.5567	7	69	2.44%
Aplasia/Hypoplasia of the middle phalanges of the toes	39	1.4074e-6	2.4077e-4	17.8770	6	2.55%	35	1.3501e-2	32.3153	3	6	1.05%
Abnormal enchondral ossification	44	1.9662e-6	2.9814e-4	6.3055	11	4.68%	53	2.1822e-2	5.9528	7	76	2.44%
Cone-shaped epiphyses of the phalanges of the hand	50	2.9126e-6	3.8865e-4	11.7523	7	2.98%	63	2.3683e-2	12.9261	4	20	1.39%
Abnormality of the wing of the ilium	53	3.2709e-6	4.1176e-4	7.7668	9	3.83%	82	4.6182e-2	7.3444	5	44	1.74%
Abnormality of the radius	55	3.5733e-6	4.3347e-4	4.2519	15	6.38%	16	1.2989e-3	6.5283	10	99	3.48%
Abnormality of the ribs	56	3.6811e-6	4.3857e-4	4.2413	15	6.38%	33	1.4125e-2	4.5837	10	141	3.48%
Deviation of the hand or of fingers of the hand	57	3.9991e-6	4.6811e-4	3.5865	18	7.66%	64	2.6620e-2	3.9170	10	165	3.48%
Radial deviation of the hand or of fingers of the hand	62	4.8814e-6	5.2530e-4	5.7246	11	4.68%	31	1.2413e-2	7.0690	7	64	2.44%
Abnormality of the epiphyses of the phalanges of the hand	64	6.5559e-6	6.8345e-4	8.4058	8	3.40%	40	1.4823e-2	10.7718	5	30	1.74%

The test set of 235 genomic regions picked 287 (2%) of all 18,549 genes.
Human Phenotype has 6,672 terms covering 3,411 (18%) of all 18,549 genes, and 256,883 term - gene associations.
6,672 ontology terms (100%) were tested using an annotation count range of [1, Inf].

GREAT

Mouse-based categories

Mouse Phenotype (20+ terms)

Global controls

Table controls: Export

Shown top rows in this table: Set

Term annotation count: Min: Max: Set

Visualize this

table: select one

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
abnormal long bone morphology	8	2.8089e-8	3.3546e-5	2.3932	46	19.57%	8	9.1817e-4	2.9035	27	601	9.41%
abnormal uterus morphology	10	4.2296e-8	4.0410e-5	3.7437	24	10.21%	101	2.9091e-2	3.5370	11	201	3.83%
disproportionate dwarf	11	6.2005e-8	5.3854e-5	6.4261	14	5.96%	37	2.8400e-3	7.4934	8	69	2.79%
abnormal tail length	15	1.1933e-7	7.6008e-5	4.3609	19	8.09%	122	3.5014e-2	4.4573	8	116	2.79%
delayed endochondral bone ossification	20	2.8930e-7	1.3820e-4	5.6497	14	5.96%	24	1.9441e-3	8.3394	8	62	2.79%
increased embryo size	21	3.5922e-7	1.6343e-4	16.1615	7	2.98%	99	2.7732e-2	21.5435	3	9	1.05%
abnormal limb morphology	24	4.0997e-7	1.6320e-4	2.0502	52	22.13%	18	1.1652e-3	2.4798	32	834	11.15%
abnormal long bone epiphyseal plate proliferative zone	26	4.6680e-7	1.7153e-4	5.0273	15	6.38%	57	6.0422e-3	6.3833	8	81	2.79%
small uterus	27	4.7403e-7	1.6774e-4	5.4178	14	5.96%	29	2.5908e-3	7.8340	8	66	2.79%
abnormal appendicular skeleton morphology	28	5.1735e-7	1.7653e-4	2.0950	49	20.85%	7	8.2587e-4	2.6857	31	746	10.80%
abnormal humerus morphology	29	5.2714e-7	1.7367e-4	3.9491	19	8.09%	17	1.1468e-3	6.0764	11	117	3.83%
short limbs	31	6.6151e-7	2.0387e-4	3.7188	20	8.51%	32	2.4579e-3	5.2662	11	135	3.83%
abnormal uterus size	34	7.3734e-7	2.0719e-4	4.5353	16	6.81%	50	4.3922e-3	5.8755	9	99	3.14%
abnormal stomach glandular epithelium morphology	37	9.7430e-7	2.5158e-4	10.9199	8	3.40%	128	3.4867e-2	10.7718	4	24	1.39%
abnormal thoracic cage morphology	38	1.0304e-6	2.5907e-4	2.4052	36	15.32%	10	9.8105e-4	3.1023	24	500	8.36%
abnormal forelimb stylopod morphology	40	1.0906e-6	2.6050e-4	3.7581	19	8.09%	20	1.4725e-3	5.8274	11	122	3.83%
abnormal cartilage development	46	1.4597e-6	3.0317e-4	2.9797	25	10.64%	12	1.0252e-3	4.5302	15	214	5.23%
female infertility	47	1.6991e-6	3.4538e-4	2.8754	26	11.06%	35	2.3819e-3	3.3430	18	348	6.27%
abnormal cartilage morphology	49	1.9670e-6	3.8352e-4	2.3751	35	14.89%	1	6.4078e-4	3.7349	23	398	8.01%
abnormal basicranium morphology	52	2.4972e-6	4.5882e-4	4.1214	16	6.81%	16	1.0465e-3	6.9495	10	93	3.48%

The test set of 235 genomic regions picked 287 (2%) of all 18,549 genes.

Mouse Phenotype has 9,554 terms covering 9,592 (52%) of all 18,549 genes, and 709,451 term - gene associations.

9,554 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Mouse Phenotype Single KO (20+ terms)

Global controls

Table controls: Export

Shown top rows in this table: Set

Term annotation count: Min: Max: Set

Visualize this

table: select one

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
abnormal long bone epiphyseal plate proliferative zone	4	6.8640e-10	1.5700e-6	8.3213	15	6.38%	13	2.0555e-3	8.9146	8	58	2.79%
disproportionate dwarf	8	2.4537e-9	2.8062e-6	8.3522	14	5.96%	9	1.9632e-3	9.4008	8	55	2.79%
abnormal long bone morphology	14	5.8168e-8	3.8013e-5	2.5775	39	16.60%	20	9.5033e-3	2.8217	21	481	7.32%
abnormal thoracic cage morphology	15	8.2821e-8	5.0515e-5	2.7775	34	14.47%	5	1.5564e-3	3.3456	22	425	7.67%
abnormal cartilage development	19	1.3192e-7	6.3524e-5	3.9455	21	8.94%	11	2.2577e-3	5.3487	12	145	4.18%
abnormal long bone epiphyseal plate morphology	21	1.5758e-7	6.8654e-5	3.9027	21	8.94%	12	2.2242e-3	5.3121	12	146	4.18%
abnormal chondrocyte morphology	25	3.1048e-7	1.1362e-4	5.1950	15	6.38%	8	1.8600e-3	8.0788	9	72	3.14%
short limbs	26	3.1104e-7	1.0945e-4	4.8465	16	6.81%	36	2.1187e-2	5.6818	8	91	2.79%
abnormal appendicular skeleton morphology	27	3.1544e-7	1.0689e-4	2.2548	44	18.72%	6	1.6094e-3	2.8560	27	611	9.41%
abnormal skeleton physiology	29	3.5088e-7	1.1070e-4	2.4777	37	15.74%	7	1.6367e-3	2.9811	25	542	8.71%
abnormal bone ossification	33	5.1574e-7	1.4299e-4	2.7783	30	12.77%	2	1.7090e-3	3.8018	20	340	6.97%
abnormal epiphyseal plate morphology	37	6.8280e-7	1.6884e-4	3.5620	21	8.94%	15	4.3087e-3	4.8778	12	159	4.18%
abnormal limb morphology	39	7.4482e-7	1.7473e-4	2.1843	44	18.72%	34	1.9492e-2	2.3485	25	688	8.71%
small uterus	41	9.9329e-7	2.2165e-4	6.0696	12	5.11%	43	2.7804e-2	7.6036	6	51	2.09%
abnormal long bone hypertrophic chondrocyte zone	42	1.0042e-6	2.1875e-4	4.4277	16	6.81%	45	3.2926e-2	5.1705	8	100	2.79%
increased bone marrow adipose tissue amount	45	1.1861e-6	2.4114e-4	154.5616	3	1.28%	51	4.2799e-2	64.6307	2	2	0.70%
abnormal uterus size	48	1.4928e-6	2.8454e-4	4.9085	14	5.96%	50	3.7337e-2	5.8002	7	78	2.44%
abnormal cartilage morphology	49	1.5706e-6	2.9325e-4	2.6870	29	12.34%	3	1.5025e-3	4.0963	18	284	6.27%
abnormal skeleton development	57	2.4260e-6	3.8940e-4	2.4761	32	13.62%	10	2.4481e-3	3.3487	20	386	6.97%
respiratory failure	62	3.3191e-6	4.8977e-4	3.3399	20	8.51%	22	1.0502e-2	4.6772	11	152	3.83%

The test set of 235 genomic regions picked 287 (2%) of all 18,549 genes.

Mouse Phenotype Single KO has 9,149 terms covering 9,408 (51%) of all 18,549 genes, and 554,948 term - gene associations.

9,149 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Some limitations

For “over-representation”-based methods we’ve discussed

- Genes are effectively all treated equally
(expression fold-change, effect sizes of each gene is not considered)
- Tends to discard genes not meeting a threshold
(a somewhat arbitrary significance or fold-change cutoff is applied)
- Genes are assumed independent
(e.g., types/strengths of genes’ biological interactions are not considered)
- Pathways/gene sets are assumed independent
(though, e.g., GO terms are hierarchically ordered)

Some extensions

“Functional class scoring” (FCS) methods

- Rather than using an arbitrary cutoff for including genes and weighing all genes equally, try to use, e.g., full matrix of expression data
- One basic idea: calculate a score for each gene, then aggregate genes' scores into pathway-level scores, and test for significance
- Example: GSEA (gene set enrichment analysis)

Example: GSEA

Rank genes by effect on/correlation with phenotype.

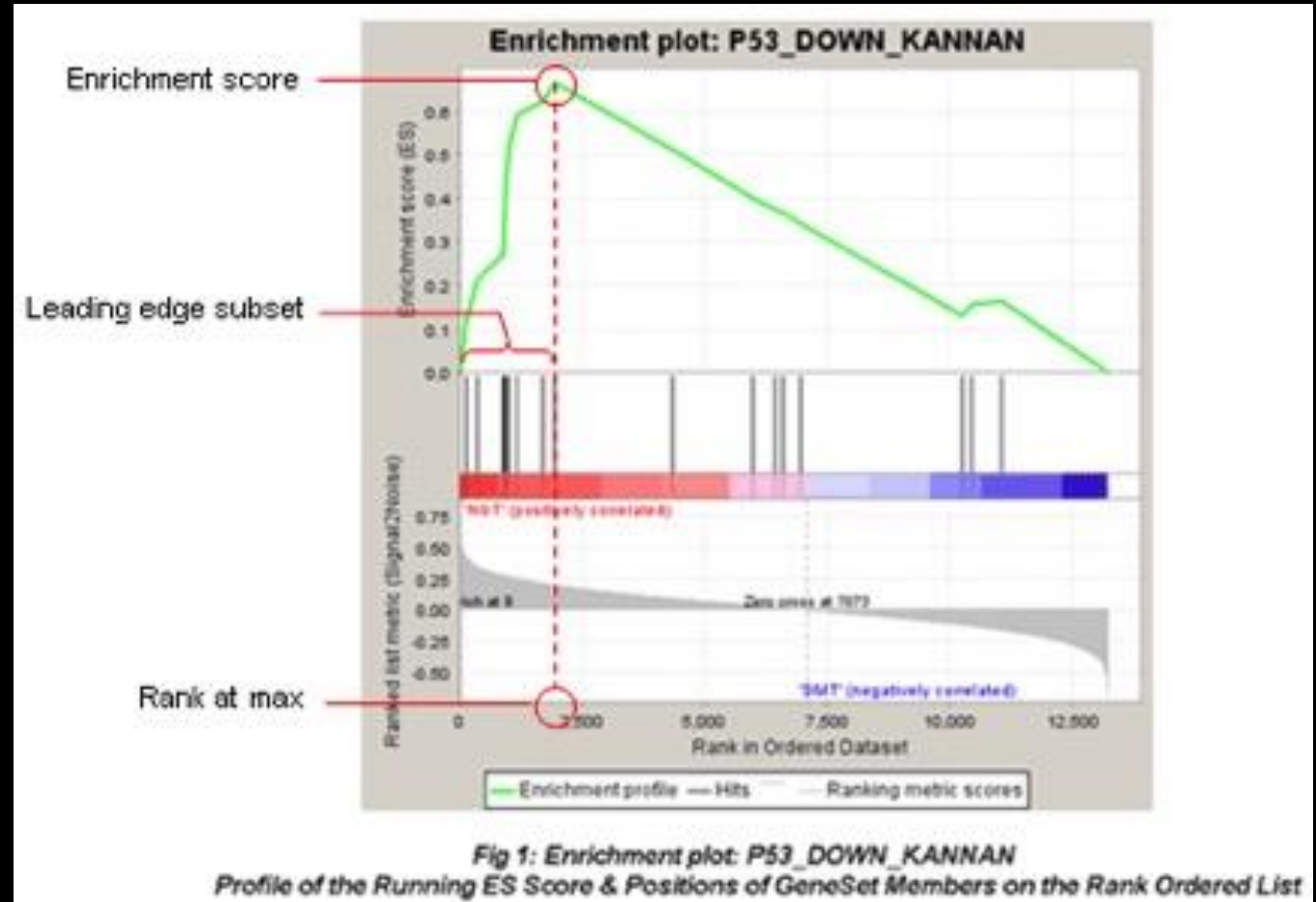
GSEA calculates the ES by walking down the ranked list of genes, increasing a running-sum statistic when a gene is in the gene set and decreasing it when it is not.

The magnitude of the increment depends on the correlation of the gene with the phenotype. The ES is the maximum deviation from zero encountered in walking the list. A positive ES indicates gene set enrichment at the top of the ranked list; a negative ES indicates gene set enrichment at the bottom of the ranked list.

The top portion of the plot shows the running ES for the gene set as the analysis walks down the ranked list. The score at the peak of the plot (the score furthest from 0.0) is the ES for the gene set. Gene sets with a distinct peak at the beginning (such as the one shown here) or end of the ranked list are generally the most interesting.

Use a permutation procedure to normalize the ES and assign each gene set an enrichment p-value. Adjust for multiple testing/try to control FDR.

(Numerous implementations exist.)



Other extensions

Pathway topology (PT)-based approaches

- ORA/FCS methods don't use "structure" of gene sets
- PT-based methods do

Review/comparison:

<https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-019-3146-1>

- Some approaches
 - Computing similarities for gene pairs in a pathway and adjusting for their "distance" in the pathway → gene score → pathway score
 - "Impact factor" approach: treat signaling pathway as a graph, using its connectivity structure to model a pathway score as a weighted sum involving differential expression of each gene in the pathway
- Some limitations: pathway connectivity could vary across cell types or with experimental conditions

Some other challenges

- Need for “higher-resolution” databases
 - Lots of alternative splicing → the “same gene” (as labeled in a pathway or gene set) could have various different meanings/contexts in a given pathway, though the pathway may have only one representation in databases
 - In an association study, how does a SNP affect associated with a phenotype affect a *pathway*?
- Missing/biased data
 - How should non protein-coding elements like pseudogenes be included? (They may be transcribed/affect pathways)
 - > 95% of GO term assertions (as of 2007) are inferred from electronic annotations (and more likely lower in quality or incorrect)
 - Lots more “manual” annotation needed

Some other challenges (con't.)

- Condition- and cell-specificity
 - Databases are built from experiments on certain cell types in certain tissues and may not generalize; this context may be unavailable.
 - Different “variations” of the “same” pathway from different contexts may erroneously lead to two genes that are actually independent to be grouped together in a gene set; this can inflate the number of genes in a pathway and lead to spurious GSA associations
 - “Canonical” pathways may differ from disease- or phenotypic-specific pathways

Some other challenges (con't.)

- Benchmarks
 - Lack of a “gold standard” since “the full biology” is generally not known; simulations can’t answer everything.

When to use each approach

- If you have GWAS results
 - Often converted to a gene list for ORA
 - Use enrichment tools that test many gene sets at once
 - FUMA
 - Enrichr
 - clusterProfiler (ORA)
 - DAVID
 - If working with genomic regions (not just genes), use GREAT
 - If you have gene-level stats, use rank-based methods (e.g., MAGMA in FUMA)
- If you have differential expression results
 - Usually you have a ranked gene list with effect sizes → use GSEA
 - Use GSEA or R packages (clusterProfiler [GSEA] or fgsea)

A note on interpreting results

Multiple testing problem: we usually test many, many gene sets

- FDR = false discovery rate = proportion of significance results that are false positives
- E.g., 20 significant pathways at FDR=0.05 $\rightarrow$ 1/20 expected to be a false positive
- FDR < 0.05 somewhat stringent; FDR < 0.1 often used for GSEA

q-value = minimum FDR at which a test is significant

- “If I call this significant, what fraction of my discoveries would be false?”
- E.g.: pathway has $p=0.0001$, $q=0.03$ $\rightarrow$ Including it among significance results means you expect 3% of significant findings are false positives
- Using FDR < 0.05 $\rightarrow$ select all results with $q < 0.05$
- Think of q-values as adjusted p-values, controlling for FDR